

Certified Defender Move Restriction in Hexo Threat Search

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Abstract

Threat-space search in Hexo has large defender branching because a legal reply may be far from the current tactics. We give finite certificates that restrict those replies without changing a verified attacker win. The main result is a horizon-parameterized dismissal theorem with ranked obligation guards and separate completion guards. Exact-copy forcing gates permit a target-specific debit, while sparse LOSS witnesses, a sharp pairing threshold, and potential-function bounds describe complementary parts of the search. The paper states the verifier contracts, records their proved limits, and separates machine-checked finite claims from mathematical proofs.

Contents

1	Introduction and reader’s guide	2
1.1	How to read the paper	3
2	The game and formal model	4
2.1	Informal rules	4
2.2	Formal model	4
3	Locality and elementary geometry	6
3.1	The three causal channels	7
4	Certificates, obligations, clocks, and coupling	9
4.1	Exact certificate grammar	9
4.2	A certificate used throughout the paper	9
4.3	Obligations and local clocks	10
4.4	The ranked zone	12
4.5	The real-ghost coupling	13
5	Main theorem: dismissal soundness	14
6	Computable zones and local reductions	15
6.1	The ranked zone theorem	15
6.2	The full-clock calculation in the running example	16
6.3	A short static cover	17
6.4	The extendable-hit kernel	17
6.5	Ranked relevance closure	18
6.6	Branch-indexed substitution	18
6.7	Finite certificate DAGs	20
6.8	Local domination patterns	20
7	The forced-hit gate calculus	21
7.1	Why pressure alone cannot be debited	22
7.2	Protected exact-copy gates	22
7.3	Transfer at a gate	24
7.4	The running certificate fires its gate	26
7.5	What remains open	26
8	Sparse LOSS witnesses	27
8.1	The sharp three-window configuration	28
8.2	The sharp five-window configuration	28

9	Pairing on the hex lattice	29
9.1	The length-six obstruction	29
9.2	The threshold construction	30
9.3	Finite verification of the periodic search	32
10	The potential layer and its boundary	32
10.1	Fixed target families	32
10.2	Birth mass and permanent local certificates	34
10.3	Finite-horizon consequences	35
10.4	How narrow is the raw threshold?	37
10.5	Dynamic greedy can be forced to lose	37
10.6	Clean escape and failure of raw renewal	39
10.7	Two static repair attempts	40
10.8	Finite reduction and the remaining boundary	40
11	Mechanized validation	41
11.1	Search-harness runs	42
11.2	Domination spot checks	42
11.3	Four finite-check campaigns	42
11.4	Validation boundary	43
12	Tightness: the limit map	44
12.1	Obligation and completion boundaries	44
12.2	LOSS and kernel boundaries	44
12.3	Clock, deadline, and grammar boundaries	44
12.4	Static coverage and the forced-hit boundary	46
13	Conclusions	46
13.1	What this paper establishes, in plain language	46
13.2	Results and limits	46
13.3	Implications for solvers	48
13.4	Future work	48
A	Notation and source crosswalk	49
A.1	Notation	50
A.2	Paper number to source tag	52

1 Introduction and reader's guide

Hexo is played on an unbounded hexagonal grid. A player wins by occupying six consecutive cells on one of the three grid axes. After the opening move, each turn normally contains two placements. A move is legal when it is close to some stone already on the board. The legal region therefore grows with the position. A distant placement can matter later even when it does not touch a current threat.

This paper studies the defender nodes of threat-space search. Threat-space search follows forcing attacking moves and asks whether every defense still leads to a win. The natural OR nodes are often narrow because the search has chosen one attacking continuation. The AND nodes are wide. A midgame node can have hundreds of legal defender replies. Testing all of them at every depth can dominate the search even when the tactical line itself is short. This is the main branching problem addressed here.

The usual local shortcut is unsafe. A defender stone outside the visible threats can expand the radius-eight legality frontier. Later stones can use that new support to reach a cell required by the attacking line. The same remote play can also start a run that completes before the planned attack. A sound restriction must account for occupancy, the windows changed by a stone, and the legality frontier created by that stone. A rule based only on the current threat cells misses two of those effects.

The remedy is a certificate. A certificate is a finite attacking strategy tree with exact positions, legal transitions, terminal witnesses, and a declared resolution time. It also records which future cells must remain available. A separate verifier checks the tree. Search may use aggressive move ordering or incomplete candidate generation, but a claimed win is accepted only when the certificate covers every omitted defender reply. This separates fast discovery from trusted verification.

The central result gives a sound coverage test for each ordinary defender node. Its mandatory zone has four parts. Direct obligations protect cells that are already legal. Ranked seed guards stop a remote reply from opening a chain into an obligation before its deadline. Touched-window guards protect windows that already contain a defender stone

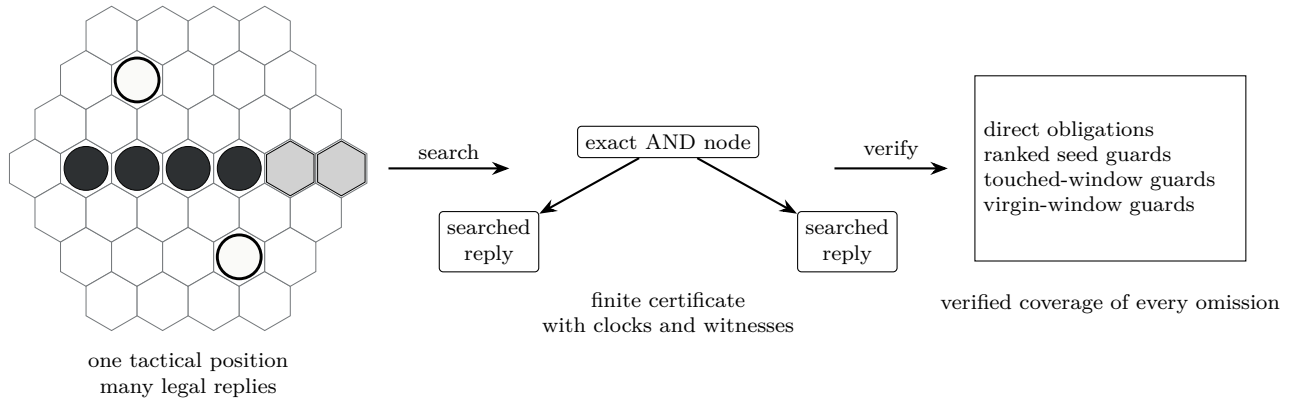


Figure 1: Orientation. Search proposes a finite attacking certificate. Verification reconstructs the obligations and clocks that justify the omitted defender replies.

and can complete in time. Virgin-window guards cover a legal seed that can reach and fill an empty window before the attack resolves. Each obligation and each window uses a local clock. A deeper certificate can therefore protect more territory than a shallow certificate, while unrelated long branches do not inflate every node.

The proof compares a real play with a ghost walk through the certificate. A searched defender reply is copied. An omitted reply is placed only on the real board, while a searched filler is placed on the ghost board. The two positions can then differ in defender stones but continue to share attacker stones, ply, mover, and budget. The four-part zone prevents those differences from occupying a required attacker cell or completing a defender window before the mapped resolution. Terminal witnesses then transfer from the ghost path to the real play.

Some forcing nodes permit a smaller hazard count. If the current threats force the defender to choose from a tight hitting kernel, an exact copied hit does not create a real-only legality frontier. The gate calculus debits that hit from the role clock and charges it to a particular defender window only when the hit lies in that window. An ignored gate keeps the full cost of the remaining defender turn, because the defender may try to win before the attacker uses a surviving threat. This is a proved, target-specific debit. It does not reduce the general path budget, assign zero cost to a substituted reply, or guarantee that every final searched set becomes smaller.

Several finite results sharpen the certificate format. A LOSS leaf needs at most three threat windows when one defender placement remains and at most five when two remain. Both caps are attained by Hexo configurations. A separate kernel theorem handles fully forced regions entered with identical real and ghost positions. Transition envelopes allow a dismissal to select a particular searched child, but they must include the current transition. Finite acyclic certificate graphs are sound when shared labels and clocks are consistent.

The pairing result locates an exact lattice threshold. No position-independent matching can put a defensive pair inside every length-six Hexo window. The density obstruction becomes equality at length seven, where an index-twelve periodic perfect matching covers every window exactly once. The construction is minimal among periodic constructions. This closes the global static pairing question for the Hexo board while leaving position-dependent local pairings available as search devices.

The potential layer describes a different defensive certificate. A weighted sum over attacker-touched live windows can block a fixed family forever when the defender moves first and the sum is below one. The same scalar applied to the changing family protects a finite horizon, not an unrestricted game. Newly touched windows add mass. An exhaustive 124-branch calculation shows that every dynamic greedy tie-breaking rule can lose from a position below the threshold. A short clean-escape proof also shows that no defense can preserve that raw sublevel set at every turn. Finite-horizon extensions and a finite reduction remain valid. The existence of a different non-greedy forever defense is open.

Local domination gives three additional solver rules. A dead-cell reply can be dismissed when its named substitute has no new legality frontier. Two hits of one threat can be interchangeable under dead-spoke and support conditions. Two placements of the same turn commute when both were legal at turn start and neither first placement wins. These rules are independent of the ranked zone and can reduce branching before certificate verification.

The limit map separates absolute game constants from bounds that are final only inside a named certificate or potential framework. It records which thresholds are attained, which are merely sufficient, and which questions remain open. This prevents a local verifier bound from being read as a position-independent law of the game.

1.1 How to read the paper

The formal development starts in [Section 2](#). The companion directory contains the longer derivations and the records of finite checks [1]. The paper gives complete proofs for the dismissal theorem, the ranked-zone reading, the gate soundness theorem, the sparse witness theorem, the pairing threshold, and the short boundary results proved here. Other supporting statements are given with proof sketches and a companion citation.

Purpose	Main route	Optional material on a first pass
Solver implementation	Sections 2, 4 to 7, 11 and 13	The density proof in Section 9 and the detailed potential boundary in Section 10.
Combinatorial results	Sections 3, 8 to 10 and 12	The verifier data structures in Sections 4 and 6.
General orientation	This section, Sections 2, 5, 7 and 13	Proof details in Sections 4 and 10 and the crosswalk in Appendix A.

Table 1: Reader routes through the paper.

Threat-space search and df-pn supply the algorithmic setting [2, 3]. The conclusions return to implementation and state which tests are safe, which reductions are conditional on extra certificate data, and which questions remain heuristic.

2 The game and formal model

2.1 Informal rules

Hexo is a two-player placement game on an unbounded hexagonal grid. The opening player places one stone at the origin. Thereafter the players alternate turns of two placements. Stones are permanent. A placement is legal on any empty cell within hex distance 8 of a stone of either player. The first player to occupy six consecutive cells on one of the three grid axes wins. A win is checked after each placement, so a win on the first placement ends the game before the second placement of that turn.

There are no captures or passes. The engine has no normal draw outcome. An unfinished play can continue on the unbounded board, but every certificate in this paper has a finite declared horizon. The informal rules and the formal model agree with the repository rule implementation [1].

We use attacker and defender as fixed roles, denoted by A and D. A ply is one placement, while a normal turn contains two plies. Everything below concerns positions after the Opening placement (the origin is occupied; budgets follow the turn structure of Definition 2.5).

2.2 Formal model

The first definition fixes the infinite board and the metric used by both the line geometry and the legality rule.

Definition 2.1 (Board, cells, and distance). The cells are the integer axial lattice $\mathbb{Z}^2 \ni x = (q, r)$. For $x = (q, r)$ and $y = (q', r')$, set

$$d(x, y) = \max\{|q - q'|, |r - r'|, |(q - q') + (r - r')|\}.$$

In plain terms. The coordinate pair names one hexagon. The third quantity in the maximum is the missing cube-coordinate difference. Distance eight is therefore an actual hex-grid radius rather than a rectangular coordinate bound.

The winning patterns are finite segments along the three line directions of the axial lattice.

Definition 2.2 (Axes and windows). The axis vectors are

$$\mathbf{a} \in \{(1, 0), (0, 1), (1, -1)\}.$$

A *window* is a set

$$W = \{s + i\mathbf{a} : 0 \leq i \leq 5\}$$

for a start cell s and an axis vector $\mathbf{a}$. Every window has exactly six cells, and every cell lies in exactly 18 windows, from three axes and six possible offsets on each axis.

In plain terms. A winning line is always one of these six-cell sets. A cell has six possible positions inside a window on each of three axes. This is why a placement changes exactly eighteen window records.

The next fact records the metric consequence of belonging to one window.

Fact 2.3 (Window collinearity). Two distinct cells of a window differ by a nonzero multiple of the window's axis vector. Thus any two cells contained in a common window are collinear along that axis and have distance at most 5.

In plain terms. Window cells lie on one grid line and are never more than five steps apart. This small diameter makes threat cells legal and supplies the locality bounds used later.

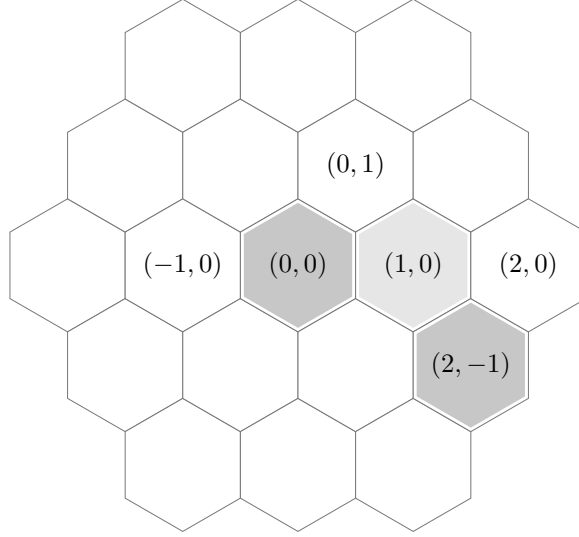


Figure 2: Pointy-top axial coordinates. The shaded cells show a shortest path from $(0, 0)$ to $(2, -1)$, and $d = \max(|2|, |-1|, |1|) = 2$.

Figure 2 shows the coordinate convention and one distance calculation.

A position records the finite board state and the exact placement remaining in the current turn.

Definition 2.4 (Positions). A position is $P = (\sigma, m, b)$, where σ is a finite partial map $\mathbb{Z}^2 \rightarrow \{A, D\}$, the mover is $m \in \{A, D\}$, and $b \in \{1, 2\}$ is the mover's number of remaining placements in the current turn. Stones are permanent, and $\text{Stones}(P) = \text{dom } \sigma$. For $X \in \{A, D\}$, let $\bar{X}$ denote the other player. For such an X and a window W , define

$$\text{cnt}_X(W, P) = |\{c \in W : \sigma(c) = X\}|.$$

When the position is clear from context, we suppress it and write $\text{cnt}_X(W)$.

In plain terms. The state records one placement phase, not only the player whose turn it is. This distinction is needed because the remaining budget can be one or two and a win can stop the turn between those placements.

Legality is local to the existing stones, while a transition must also account for immediate termination and the two-placement turn structure.

Definition 2.5 (Legality and transitions). A cell c is *legal* in P if and only if

$$c \notin \text{dom } \sigma \quad \text{and} \quad \exists s \in \text{dom } \sigma : d(c, s) \leq 8.$$

Write $\text{Legal}(P)$ for the set of cells legal in P . The mover places on a legal cell c , producing $\sigma' = \sigma \cup \{c \mapsto m\}$. If some window W then satisfies $\text{cnt}_m(W) = 6$, the game terminates immediately and m wins. Wins are therefore checked after each placement and may occur on the first placement of a two-placement turn. If there is no win, the next position is $(\sigma', m, 1)$ when $b = 2$, and $(\sigma', \bar{m}, 2)$ when $b = 1$.

In plain terms. Any existing stone of either colour can justify legality. A remote defender reply can therefore create legal cells for both players. The soundness proof must track that frontier even when no current threat window is touched.

Window states distinguish usable winning lines, blocked lines, and lines that can be completed within one normal turn.

Definition 2.6 (Window states). A window W is *alive* for X if $\text{cnt}_{\bar{X}}(W) = 0$. An all-empty window is alive for both players. It is *dead* if $\text{cnt}_A(W) \geq 1$ and $\text{cnt}_D(W) \geq 1$. It is an *X-threat* if it is alive for X and $\text{cnt}_X(W) \geq 4$.

In plain terms. An alive window can still become a win for the named player. A dead window cannot, because stones are permanent. Four stones are a threat because the two remaining cells fit in one normal turn.

The three panels in Fig. 3 give the basic mask patterns used by this terminology.

The implemented one-ply analysis tests whether the mover can win in the remaining placements or whether the opponent's threats exceed the defensive budget.



Figure 3: Alive, dead, and threat window states. Filled disks are attacker stones and open disks are defender stones.

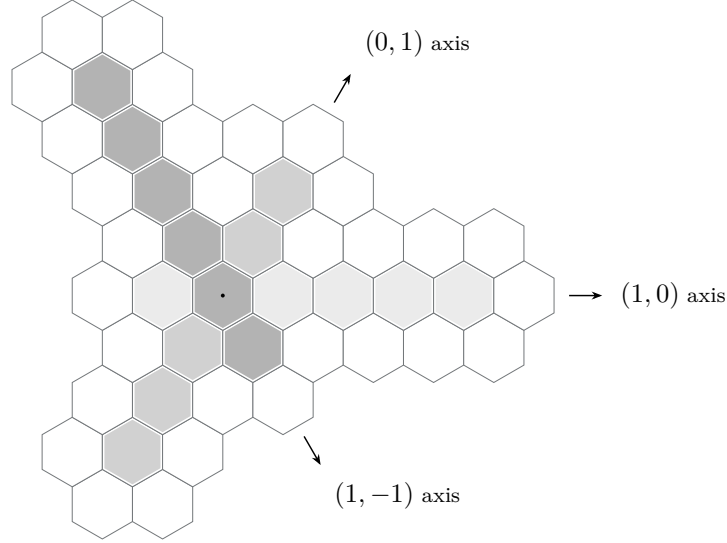


Figure 4: One six-cell window on each axis through the central cell, shown with three fill shades. Sliding a window so that the cell occupies offsets 0 through 5 gives 3 axes $\times$ 6 offsets = 18 windows through the marked cell.

Definition 2.7 (λ^1 predicates). For mover m with budget b , $\text{own_win_now}(P)$ holds if and only if there is an m -alive window W such that

$$\text{cnt}_m(W) = 5, \quad \text{or} \quad \text{cnt}_m(W) = 4 \text{ and } b = 2.$$

The value $\text{hit}(P)$ is the multiset of empty-cell sets of the $\bar{m}$ -threat windows. The value $\text{mhs}(P)$ is the minimum number of cells meeting every set in $\text{hit}(P)$, computed exhaustively for values at most 2. The λ^1 verdict is WIN when $\text{own_win_now}(P)$ holds, LOSS when it does not hold and $\text{mhs}(P) > b$, and undecided otherwise. The post-opening soundness of these implemented verdicts is assumed here.

In plain terms. The tactical test reduces a family of threat empties to a hitting-set question. A defender budget of two can answer a family with transversal number at most two. The LOSS result also requires the mover to lack a faster win of its own.

Absolute placement indices allow later certificates to state an exact win horizon without treating a two-placement turn as simultaneous.

Definition 2.8 (Plies and horizon). A *ply* is one placement, and ply indices are absolute along a play. For a position P with mover m , the *defender-placement count before ply T* , written $\mathfrak{D}(P, T)$, is the number of plies strictly before T at which D places, as determined by (m, b) and alternation.

In plain terms. Certificate deadlines count placements, not turns. This convention records the exact order in which a first-stone win, a second-stone win, and the next turn can occur.

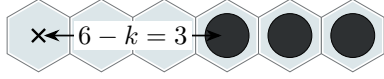
The final convention fixes the direction needed when defender replies are removed from an attacker-win certificate.

Definition 2.9 (Dominance direction). Outcomes are evaluated for the attacker A, with $\text{WIN} \succ \text{nonWIN}$. A *sound dismissal* of a defender reply c at a node N means that, if the certificate proves that A wins against all searched replies, then A also wins against c .

In plain terms. The paper removes defender replies only while proving an attacker win. A dismissal is sound when the omitted reply cannot turn that verified win into a nonwin. No symmetric dominance claim is intended.

3 Locality and elementary geometry

The certificate proof uses two kinds of locality. A six-cell window has small diameter, while a new stone changes legality only within radius eight. The first group of statements controls tactics inside one window. The second group lists every channel that must be checked when two positions differ. Complete derivations are recorded in the companion proof [1].



(a) The equality case for $k = 3$.



(b) The threat case for $k = 4$.

Figure 5: The packing bound and its tight configurations. A cross marks the empty cell at offset zero.

A line segment with many stones has few empty cells. The farthest empty cannot be farther from all stones than the number of empties permits.

Lemma 3.1 (Packing bound). *In a window with $k \geq 1$ stones of one colour and no stones of the other, every empty cell is within distance $6 - k$ of a same-window stone. The bound is tight.*

Proof sketch. Index the window cells from 0 to 5 along its axis. If an empty cell has nearest in-window stone at axis distance δ , then that cell and the $\delta - 1$ cells between them are empty. Hence $\delta \leq 6 - k$. Axis distance equals hex distance inside a window by Fact 2.3. Stones at offsets $6 - k, \dots, 5$ and an empty at offset 0 attain equality. $\square$

In plain terms. Four stones in a live window put every remaining cell within distance two. Three stones give distance three. These two substitutions drive most of the local tactical bounds.

Whenever a window contains at least one stone and at least one empty, the boundary between occupied and empty offsets gives an adjacent pair.

Lemma 3.2 (Boundary pair). *Any window containing at least one stone and at least one empty cell has an empty cell at distance one from a same-window stone.*

Proof sketch. The occupied offsets form a nonempty proper subset of a six-cell line. At a boundary of this subset, an occupied offset and an empty offset are adjacent. Their axis distance and hex distance are one. $\square$

The packing bound places every current tactical answer close to a stone that already supports the same window.

Theorem 3.3 (Single-turn tactical locality). *Every placement that completes a win within the current turn, and every empty cell of every threat window, lies within distance two of an existing stone.*

Proof sketch. A completion using at most two placements starts from an alive window with at least four stones of that player. A threat has the same count lower bound by Definition 2.6. Apply Lemma 3.1 with $k = 4$. $\square$

In plain terms. Immediate wins and direct threat answers are always legal after the opening. They lie inside the smaller radius-two neighborhood, even though the game allows radius-eight placements.

Creating a new threat needs three earlier stones in its witness window. This gives the radius used by the common local candidate generator.

Theorem 3.4 (Threat-creation locality). *A placement that creates an X -threat lies within distance three of an existing X stone. The bound is tight.*

Proof sketch. Immediately before the placement, its witness window contains at least three X stones. Apply Lemma 3.1 with $k = 3$. Stones at offsets 3, 4, 5 and a placement at offset 0 attain distance three. $\square$

In plain terms. Radius three contains every move that turns an existing three-stone support into a threat. It does not contain every setup move in a long certificate. A fourth future setup placement can enter a window that is empty at the current node.

3.1 The three causal channels

A transfer argument compares positions that differ by one or more defender stones. The next lemma gives a complete checklist for the one-stone step.

Lemma 3.5 (Difference accounting). *Let $P' = P \pm \{x\}$ differ from P by one stone at cell x . Every game-relevant predicate difference is confined to the following channels:*

- (C1) *occupancy and ownership of x ;*
- (C2) *the masks of the eighteen windows containing x , including every completion, alive/dead, threat, own_win_now, hit, and mhs predicate;*
- (C3) *the legality frontier $\{c : d(c, x) \leq 8\} \setminus \text{dom } \sigma$.*

Proof sketch. The transition rule reads occupancy at the move cell, the legality predicate, and incident window masks. All tactical predicates are functions of window masks. A stone changes only its eighteen incident masks and supplies legality only inside its radius-eight ball. The position has no other mutable state. $\square$

In plain terms. Matching threat counts is not enough. A valid simulation must also preserve the cells needed later and explain any new legal cells. The main coupling and the domination patterns use this same checklist.

Stones never leave the board. This gives the direction in which window states can change along a play.

Lemma 3.6 (Permanence). *Window masks are coordinatewise nondecreasing along every play. Dead windows remain dead. A window alive for X at ply t is alive for X at every earlier ply.*

Proof sketch. Each transition adds one stone and removes none. A mixed-colour mask cannot lose either colour. If an opponent stone had occupied a window earlier, it would still occupy it at ply t . $\square$

Adding defender stones away from named cells and windows is safe in a particular direction. The exact completion clause prevents an invalid monotonicity shortcut.

Lemma 3.7 (Inert-extension transfer). *Let $P' = P + X$, where X is a finite set of defender stones. Let $\mathcal{W}$ be a set of windows and C a set of cells such that no $x \in X$ lies in a member of $\mathcal{W}$ and $X \cap C = \emptyset$. Then:*

- (a) *every cell of C empty in P is empty in P' ;*
- (b) *every $W \in \mathcal{W}$ has identical masks in P and P' ;*
- (c) *$\text{Legal}(P) \setminus X \subseteq \text{Legal}(P')$, and every cell of C legal in P is legal in P' ;*
- (d) *the A-complete windows of P' are exactly the A-complete windows of P that avoid X .*

Thus an attacker completion on a window known to avoid X transfers from P to P' , and every attacker completion in P' was available in P .

Proof sketch. The first two parts are occupancy and mask equality. Added stones can add legality support, while their occupied cells leave the legal set. A window completed by the attacker must contain no defender stone, so exactly the X -free completions remain. $\square$

In plain terms. Extra defender stones can make more moves legal, but they can also block an attacker window. Transfer is valid only for named cells and windows that avoid those stones.

Deleting defender stones helps attacker masks but can remove the only support that made a later attacker placement legal.

Lemma 3.8 (Deletion monotonicity and well-formed legality). *Let $P'' = P - Y$, where Y is a finite set of defender stones of P . Then:*

- (a) *every attacker threat of P is an attacker threat of P'' , so the defender's $\text{mhs}(P'') \geq \text{mhs}(P)$ and defender λ^1 LOSS verdicts transfer from P to P'' ;*
- (b) *attacker-complete windows of P are attacker-complete in P'' ;*
- (c) *a cell empty in P is empty or newly empty in P'' ;*
- (d) *legality may shrink. Every relied-upon attacker placement must be within distance eight of a stone outside Y . In the uses below it is within distance three of an attacker stone, as in Theorem 3.4, or within distance eight of an unaffected root stone.*

Proof sketch. Deleting defender stones cannot kill an attacker threat or completion. It can only make occupied cells empty. It can also delete a legality witness, which is why the last condition is independent of mask monotonicity. $\square$

In plain terms. Removing defender stones makes the attack no weaker inside existing windows. It can still make an attacker move illegal. Every deletion argument must name an attacker or root stone that continues to justify that move.

The final elementary count separates stones already in a window from defender placements that remain before a deadline.

Lemma 3.9 (Completion counting). *If $\text{cnt}_D(W, P) = s$ and the defender makes at most D placements at plies strictly before T , then W cannot become defender-complete before T unless $s + D \geq 6$.*

Proof sketch. A defender completion contains six defender stones. At most s are present at P , and at most D further stones can be placed before T . $\square$

These lemmas do not by themselves define a safe reply set. The certificate must say which cells and windows are relevant, and it must attach a deadline to each one. That data is introduced next.

4 Certificates, obligations, clocks, and coupling

A restricted defender search is useful only when a verifier can reconstruct why each omitted reply is harmless. The certificate therefore records exact game states, exact searched successors, the cells needed by future attacker moves, and separate clocks for occupancy and completion hazards. A second play, called the ghost play, follows the recorded tree while the real defender may choose an omitted reply.

4.1 Exact certificate grammar

The grammar fixes the data that a verifier receives. In particular, a leaf is not a bare tactical verdict. It contains the windows and deadlines needed to replay that verdict after real and ghost defender stones have diverged.

Definition 4.1 (Certificate). A *certificate* $\mathcal{C}$ for the assertion that A wins from P_0 is a finite rooted tree. Every node is labelled by an exact position in the sense of Definition 2.4, including its mover and remaining placement budget. The root is the nonterminal position P_0 . The verifier recomputes the ply clock as the root ply plus path depth. The *leaf-ply* is the index of the last placement on the path into a leaf. Every edge is the exact legal successor specified by Definition 2.5. No edge is defender-terminal.

An *OR-COMPLETION leaf* has A to place. It names one designated placement, an A-complete witness window for that placement, and the completion ply.

A *WIN leaf* has A to place and passes a verifier check of `own_win_now`. It names an A-alive count-5 witness for either budget, or an A-alive count-4 witness when the attacker budget is 2. Its declared resolution lies in the current attacker turn.

A *LOSS leaf* has D to place with budget b . It names a family $\mathcal{T}$ of A-threat windows whose empty-set family has transversal number $\tau(\mathcal{T}) > b$, and it passes the defender check $\neg \text{own_win_now}$. For every complete nonterminal remainder H of the defender turn, consisting of exactly b placements unless the game ends earlier, some $W_H \in \mathcal{T}$ satisfies

$$E(W_H, P_L) \cap H = \emptyset.$$

The attacker then fills the at most two remaining empties of W_H during the following turn. The declared worst-case resolution is

$$\text{leaf-ply} + b + 2.$$

The verifier may require at most three named windows when $b = 1$ and at most five when $b = 2$, by Lemma 8.1.

An internal OR node has one designated attacker placement and its exact child. An internal AND node N has an independently nonempty set $S(N)$ of searched legal defender replies and one exact child for each member of $S(N)$. Every other legal defender reply carries a dismissal claim. The verifier retains the internal defender check $\neg \text{own_win_now}$ as a diagnostic. It follows from the ordinary completion zone by Lemma 6.8; it remains mandatory at LOSS leaves and is an explicit premise of Theorem 6.3. Every terminal resolution is finite. The global horizon T is the maximum of the declared resolutions on all root-to-leaf paths. The finite DAG form is given in Definition 6.9.

In plain terms. The certificate is an exact, finite claim about one attacker strategy. OR nodes state what the attacker will do. AND nodes retain enough defender replies to justify every omitted one. The typed leaves prevent a verifier from accepting a tactical label without the cells and time needed to realize it.

4.2 A certificate used throughout the paper

One finite certificate supplies the numerical example in this and the next three sections. Its root has D to place with budget 1. There are fifteen stones of each colour:

$$\begin{aligned} A_0 &= \{(q, 0) : 0 \leq q \leq 4\} \cup \{(0, -7)\} \\ &\quad \cup \{(-3, 20), (-2, 20), (-1, 20), (0, 17), (0, 18), (0, 19), \\ &\quad \quad (-3, 23), (-2, 22), (-1, 21)\}, \\ D_0 &= \{(-1, 0), (8, 0), (8, 8), (8, 16), (8, 24), (8, 32), \\ &\quad (0, 40), (1, 40), (2, 40), (-8, 0), (0, -8), (-8, 8), \\ &\quad (8, -8), (16, -8), (16, 0)\}. \end{aligned}$$

The support chain makes every displayed module legal under the radius-8 rule. Enumeration of the 18 incident windows per stone finds no complete window and finds at most three defender stones in any defender-alive window [1].

The root has the singleton attacker threat

$$T = \{(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (5, 0)\}, \quad h = (5, 0).$$

Its empty-set family is $\{\{h\}\}$. The exact searched reply is h . The attacker then places at $(3, 20)$ and $(0, 20)$ and reaches a defender budget-2 LOSS leaf. The three witness empty sets there are

$$\{(1, 20), (2, 20)\}, \quad \{(0, 21), (0, 22)\}, \quad \{(1, 19), (2, 18)\}.$$

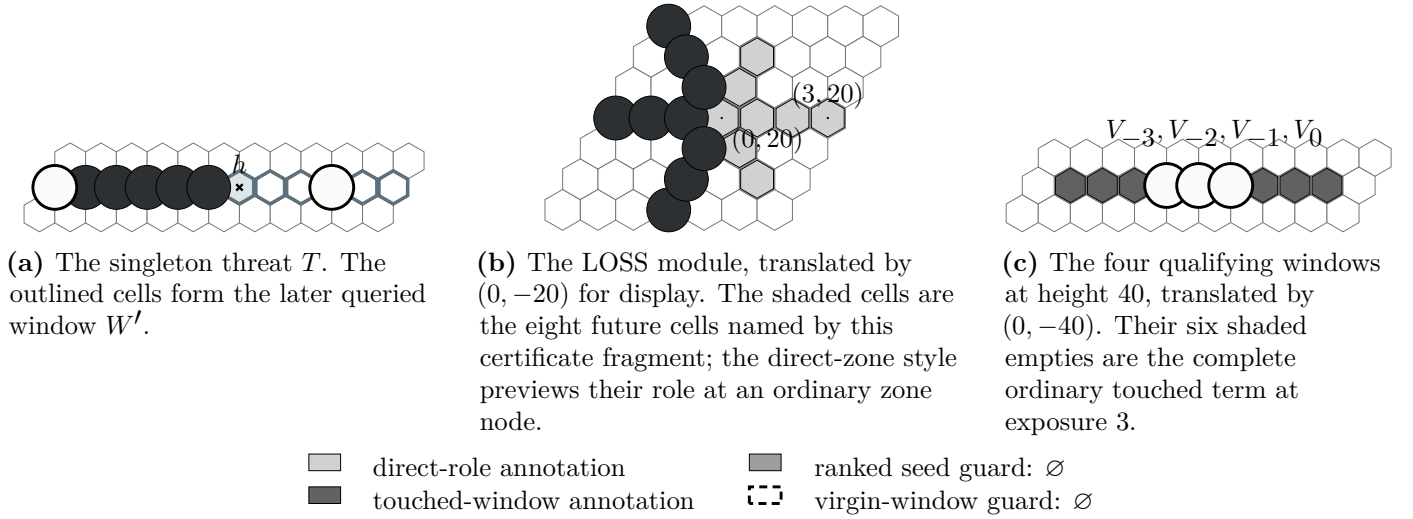


Figure 6: Tactical modules of the 15–15 running position. Blank axial distance between the modules is suppressed; the coordinate lists in the text specify the complete position. The direct and touched shadings give the exact ordinary full-clock terms; the key records that the seed and virgin terms are empty. The root is a forcing gate and instead searches the exact kernel $\{h\}$; the complete ordinary decomposition and the verified exposure calculations for W and W' appear in Section 6.2. Filled disks are attacker stones and open disks are defender stones.

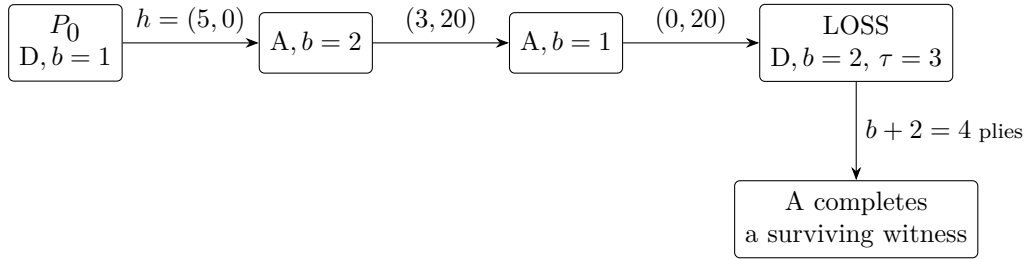


Figure 7: The certificate fragment used in the running example. The root has one exact searched reply. The two attacker OR edges lead to a LOSS leaf with three disjoint two-cell witnesses. Section 7 gives the forcing-gate interpretation of the root edge.

They are pairwise disjoint, so their transversal number is 3.

4.3 Obligations and local clocks

A global ply horizon overstates the work remaining on most certificate branches. Three local objects replace it: a defender-placement budget for a whole subtree, a deadline for each needed cell, and an exposure for each window.

Definition 4.2 (Admissible local defender budget). The exact local budget $B(N)$ is measured in defender placements and is defined by

$$\begin{aligned}
 B(\text{OR-COMPLETION}) &= B(\text{WIN}) = 0, & B(\text{LOSS}, b) &= b, \\
 B(\text{OR}) &= B(\text{child}), & B(\text{AND}) &= 1 + \max_C B(C).
 \end{aligned}$$

The two placements of a defender turn are two D4 edges. A LOSS leaf counts its remaining b placements. Exact maxima are optional: a verifier may accept any nonnegative integral labels satisfying $B(L) \geq b$ at every LOSS leaf, $B(N) \geq B(C)$ on every OR edge, and $B(N) \geq 1 + B(C)$ on every AND edge.

In plain terms. The budget counts only defender placements that can still occur on the selected branch. An attacker edge costs nothing. A defender edge costs one, and a LOSS leaf reserves enough budget for the remainder that is not expanded as certificate edges.

Only cells that the attacker may later need must be protected from a dismissed defender stone. The obligation definition records those uses and their exact deadlines.

Definition 4.3 (Compressed obligations). Write $E(W, P) = W \setminus \text{Stones}(P)$. A *live obligation role* below a node N is either a future designated certificate attacker placement, including an OR-COMPLETION placement, or a triple

(L, W, y) for a named WIN or LOSS witness W at a leaf L and a cell $y \in E(W, P_L)$. The second kind represents every permitted leaf continuation placement, including the adaptive LOSS choice.

Let $\Omega(N)$ be all such roles at reachable descendants of N . Define

$$\text{Obl}(\mathcal{C}, N) = \text{Prot}(N) = \{y : \text{some live role in } \Omega(N) \text{ is carried by } y\}.$$

An attacker-placement role remains live through the check immediately before its designated OR step. A witness-empty role remains live through the leaf entry check. Each role is discharged immediately after that check. A full witness window is not itself an obligation. Reachable-descendant nesting gives $\text{Prot}(M) \subseteq \text{Prot}(N)$ whenever M descends from N .

In plain terms. An obligation is a future use of one cell, not a permanent reservation of an entire witness window. A dismissed defender stone may alter unrelated cells without invalidating the certificate. The verifier protects each needed cell only until the moment when that use is checked.

A cell-specific clock charges only defender opportunities that occur before one named role is checked.

Definition 4.4 (Cell-specific deadlines). The deadline of a role ρ is its designated attacker placement, or leaf entry for a WIN or LOSS witness-empty role. At a node N from which ρ is reachable, $r_N(\rho)$ is an integral upper bound on the number of defender placements before that deadline. Exact ranks count the maximum number of AND edges on a reachable path. While the role is live, the verifier checks

$$r_N(\rho) \geq r_C(\rho) \quad \text{on an OR edge}, \quad r_N(\rho) \geq 1 + r_C(\rho) \quad \text{on an AND edge},$$

and checks $r = 0$ at the deadline. If several roles use one cell, set

$$r_N(y) = \max\{r_N(\rho) : \rho \in \Omega(N) \text{ is live and carried by } y\}.$$

The role is discharged only after its deadline check. A seed band is formed at an internal AND node only when $r_N(y) \geq 1$; no negative radius is formed at $r = 0$. LOSS witness empties need protection only through leaf entry, because the adaptive leaf contract controls the remaining placements. Defender-completion windows use Definitions 4.2 and 4.5, not role ranks. The uniform admissible assignment before a deadline is $r_N(\rho) = B(N)$.

A separate window clock stops when the planned attacker first enters that window, because later defender fills cannot precede that entry.

Definition 4.5 (Per-window defender exposure). For each window W , let $E_Q^D(W)$ be the maximum number of future defender placements before the certificate attacker wins or first places in W . Its exact recurrence is

$$E_Q^D(W) = \begin{cases} 0, & Q \text{ is WIN or OR-COMPLETION,} \\ b, & Q \text{ is LOSS with defender budget } b, \\ 0, & Q \text{ is an OR whose designated move enters } W, \\ E_C^D(W), & Q \text{ is any other ordinary OR,} \\ 1 + \max_C E_C^D(W), & Q \text{ is an internal AND.} \end{cases}$$

If W is already not alive for D, set $E_Q^D(W) = 0$. The attacker *enters* W when its placement makes W permanently not alive for D; the window need not become dead. Each overlapping window has its own clock. The designated stopping move in W is also an obligation, so a separate occupancy check keeps that stop playable in the real game.

In plain terms. Role ranks measure how many opportunities remain to occupy a particular needed cell. Window exposure measures how many defender stones can still enter one particular window before the attacker blocks it or wins. These are different hazards, so one scalar is not used for both.

The first few threat-creating attacker moves can be located without examining their eventual certificate nodes.

Lemma 4.6 (Short-range attacker-obligation containment). *In a certificate whose attacker placements are all threat-creating, let a placement be the attacker's k th future placement from N . If $k \leq 3$, then its threat window contains at least*

$$3 - (k - 1) \geq 1$$

attacker stones already present in P_N . Its obligation therefore lies in an attacker-touched window. From the fourth future placement onward, a setup cell may lie in a window that is virgin at N .

Proof sketch. Immediately before a threat-creating placement, its witness window has at least three attacker stones. At most $k - 1$ of them can be earlier future placements, leaving the displayed number in P_N . The fourth placement is the first for which this count may be zero. See [1]. $\square$

The recurrences provide the inequalities used when the coupling descends to a certificate child.

Lemma 4.7 (Local-clock and nesting facts). *If a descendant M is reached from N after k defender placements, then*

$$B(M) + k \leq B(N), \quad \text{cnt}_D(W, P_M) + B(M) \leq \text{cnt}_D(W, P_N) + B(N), \\ r_M(\rho) + k \leq r_N(\rho) \quad \text{for every still-live role } \rho.$$

Before the attacker enters W ,

$$E_M^D(W) + k \leq E_N^D(W), \quad \text{cnt}_D(W, P_M) + E_M^D(W) \leq \text{cnt}_D(W, P_N) + E_N^D(W).$$

Every selected path below N , including a LOSS remainder, has at most $B(N)$ defender placements before its declared attacker resolution. Also $E_N^D(W) \leq B(N)$, every exact role rank is at most $B(N)$, and $\text{Prot}(M) \subseteq \text{Prot}(N)$.

Proof sketch. Sum the verifier inequalities along the path. Each AND edge contributes one and each OR edge zero; the LOSS base covers its unexpanded remainder. A window gains at most one defender stone on each counted defender edge, and reachable roles can only disappear after descent. The D16 recurrence gives the corresponding fixed-window bounds until its stated stop. See [1]. $\square$

4.4 The ranked zone

The searched set must block three distinct failure modes. A defender may occupy a cell needed later, may approach such a cell through a chain of newly legal moves, or may complete a window before the mapped attacker blocks it. Touched and virgin windows require different completion tests.

For a set U of cells, write

$$B_s(U) = \{c : d(c, u) \leq s \text{ for some } u \in U\}, \quad d(c, W) = \min_{w \in W} d(c, w).$$

Definition 4.8 (Zone-carrying certificate). At every internal AND node N , define

$$Z_{\text{dir}}(N) = \text{Prot}(N) \cap \text{Legal}(P_N), \\ Z_{\text{seed}}(N) = \bigcup_{\substack{y \in \text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N)) \\ r_N(y) \geq 1}} (\text{Legal}(P_N) \cap B_{8(r_N(y)-1)}(\{y\})), \\ Z_{\text{touch}}(N) = \bigcup \left\{ E(W, P_N) : \begin{array}{l} W \text{ is alive for D, } \text{cnt}_D(W, P_N) \geq 1, \\ \text{cnt}_D(W, P_N) + E_N^D(W) \geq 6 \end{array} \right\}, \\ Z_{\text{virgin}}(N) = \left\{ c \in \text{Legal}(P_N) : \begin{array}{l} \text{some all-empty window } W \text{ has } E_N^D(W) \geq 6, \\ d(c, W) \leq 8(E_N^D(W) - 6) \end{array} \right\}.$$

Every cell in $Z_{\text{touch}}(N)$ is legal: Lemma 3.1 places it within distance at most 5 of an in-window defender stone.

The certificate is *zone-carrying* when $S(N)$ is independently nonempty at every internal AND node and the following verifier clauses hold.

(Z2) $S(N) \supseteq Z_{\text{dir}}(N) \cup Z_{\text{touch}}(N) \cup Z_{\text{virgin}}(N)$.

(Z4) Every certificate attacker placement, including one permitted by a leaf contract, is within distance 8 of an attacker stone in its predecessor position or a root stone unaffected by the coupling.

(Z5') $S(N) \supseteq Z_{\text{seed}}(N)$.

With the uniform assignment $r_N(y) = B(N)$, the last clause becomes

$$S(N) \supseteq \text{Legal}(P_N) \cap B_{8(B(N)-1)}(\text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N))).$$

All four components are finite and verifier-enumerable. The virgin query can be inverted from each legal candidate over the finite radius allowed by $E_N^D(W) \leq B(N)$.

In plain terms. Direct obligations are legal cells that the defender could occupy now. Seed guards stop a sequence of dismissed moves from carrying legality toward an obligation that is not yet legal. Touched and virgin guards stop the same kind of sequence from completing a defender window. Current threat-hitting cells are useful search candidates, but they are not a premise of this zone theorem.

The seed radius is set by the number of remaining opportunities to advance a radius-8 legality chain.

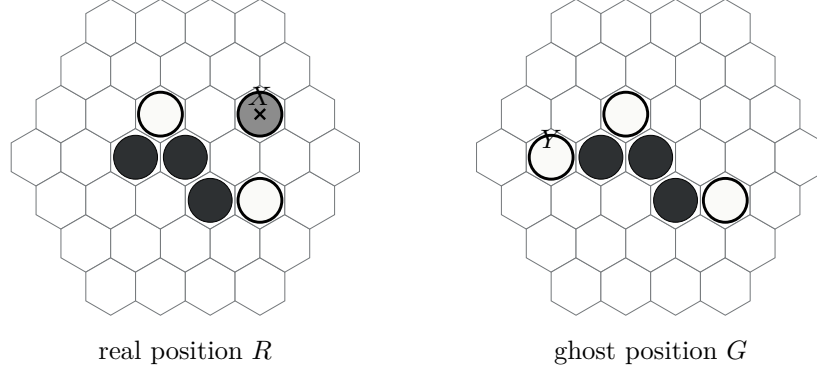


Figure 8: A schematic coupled pair. Attacker stones and the two unmarked defender stones agree. The crossed defender stone belongs only to R and hence to X ; the labelled open stone belongs only to G and hence to Y . Both boards retain the same mover, budget, and ply.

Lemma 4.9 (First protected occupation). *Suppose every direct legal protected cell is searched and every ghost-legal dismissal x at a node N lies outside radius $8(B(N) - 1)$ of*

$$\text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N)).$$

Then no defender placement creates a real-only stone in the current protected set. In the ranked form, use the maximum live role rank $r_N(y)$ for each target y in place of $B(N)$; the same conclusion holds through that role's deadline.

Proof sketch. Assume the first bad occupation is at y . If y is ghost-legal, it lies in Z_{dir} and is copied. Otherwise trace real-only legality backward to the first ghost-legal dismissed seed x_0 ; if y is the p th defender placement in that chain, then $d(x_0, y) \leq 8(p - 1)$ and nesting keeps y protected at x_0 . The local budget gives $p \leq B(N_{x_0})$, or the live role gives $p \leq r_{N_{x_0}}(y)$, so x_0 was in the required seed band. See [1]. $\square$

The distance inequality is attained for an exact rank r : a ghost-legal seed may be followed by $r - 1$ successive distance-8 defender placements, with the protected target last. This does not establish sharpness of the uniform B -only wrapper.

4.5 The real-ghost coupling

The coupling follows an exact certificate path even when the real defender plays an omitted reply. Canonical set differences record the defender stones that occur in only one of the two positions.

Definition 4.10 (Real-ghost coupling). At each coupled step, the real position R and ghost position G have the same ply, mover, and remaining budget, and have identical attacker stones. Define

$$X = \text{Stones}_D(R) \setminus \text{Stones}_D(G), \quad Y = \text{Stones}_D(G) \setminus \text{Stones}_D(R).$$

Every member of X entered at a dismissal for which the ordinary zone or a selected Definition 6.6 envelope certified avoidance of the live obligations. Every member of Y entered as a searched ghost filler. The coupling maintains

$$X \cap \text{Prot}(N) = \emptyset$$

at every visited node. A history $\hat{X} \supseteq X$ retains every cell ever added to X . For every window W the canonical identity and its useful upper bound are

$$\text{cnt}_D(W, R) = \text{cnt}_D(W, G) + |X \cap W| - |Y \cap W| \leq \text{cnt}_D(W, G) + |\hat{X} \cap W|. \quad (\text{MI})$$

In plain terms. The ghost game consumes one exact certificate edge on every defender ply. When the real defender chooses another cell, the two defender boards may differ, but their attacker board and turn state remain synchronized. The sets X and Y make every occupancy and mask discrepancy explicit.

The completion guard uses the mask inequality together with a separate causal chain for the first omitted fill of a window.

Lemma 4.11 (Split completion safety). *In the coupling of Definition 4.10, under Items (Z2) and (Z4) and clause (Z5'), no real defender play completes a fixed window W before the mapped certificate attacker wins or first enters W .*

Proof sketch. If no cell of W was dismissed, (MI) gives the ghost at least the real defender count, contradicting the ban on defender-terminal certificate edges. If the first omitted fill enters a touched ghost window, the $\text{cnt}_D + E^D \geq 6$ test places that fill in Z_{touch} . If the ghost window is virgin, a ghost-legal first fill lies in Z_{virgin} directly; a ghost-illegal fill traces back through j real-only legality links to a seed at distance at most $8j$, while completion requires exposure at least $j + 6$. The same count includes every LOSS remainder, and the mandatory leaf $\neg \text{own_win_now}$ check bounds a no-dismissal remainder by five stones. See [1]. $\square$

For one fixed window the virgin arithmetic is attained at exposure 7: a legal seed at distance 8 may be followed by the first window fill and the five remaining fills. This fixed-window trace does not establish sharpness of the complete union Z_{virgin} .

5 Main theorem: dismissal soundness

The certificate controls a ghost play, not the defender's actual choices. The proof compiles its designated OR moves into a real attacker strategy and maps every real defender placement to one certificate AND edge. Direct obligations preserve attacker moves, ranked seed guards preserve future occupancy, and the completion guards prevent a defender win during the mapping.

Theorem 5.1 (Local-clock, pathwise dismissal soundness). Let $\mathcal{C}$ obey Definitions 4.1 to 4.5, 4.8 and 4.10 and be zone-carrying under Items (Z2) and (Z4) and clause (Z5'). For every real defender play against the compiled attacker strategy, either the real attacker wins strictly earlier than the mapped ghost resolution, or the play maps to a finite certificate path and A completes by that path's declared resolution. Consequently A wins by the global maximum T , and every certified dismissal is sound.

In plain terms. The real defender may play any legal cell, including one omitted from the certificate. The ghost consumes a searched reply on the same ply and keeps following the finite certificate. The theorem says that the resulting board differences cannot remove a required attacker move or create a defender win before the recorded attacker resolution.

Proof. Couple the real position R , initially P_0 , to a ghost walk G on $\mathcal{C}$. Maintain equal ply, mover, and budget, identical attacker stones, and the canonical sets X , Y , and $\hat{X}$ from Definition 4.10. Process a typed leaf immediately when the ghost walk reaches it.

Step O: an OR node with designated placement c . The live placement role puts c in Prot at every current ancestor. The invariant $X \cap \text{Prot} = \emptyset$ gives $c \notin X$. The ghost position leaves c empty, so $c \notin Y$: a cell in Y is ghost-occupied. It follows that c is empty in the real position. Clause Item (Z4) supplies an attacker stone or unaffected root stone that is a legality witness in both positions. The compiled strategy therefore places c in both games. Their attacker-stone sets remain identical.

Suppose the OR step is an OR-COMPLETION leaf. Immediately before c , the other five cells of its named window are shared attacker stones. The window is attacker-complete in the ghost after c , so it had no ghost defender stone before the move and contains no Y -cell. The cell c is not in X . The same placement therefore completes the same real window at the declared ply. If the real placement completes another window that is not complete in the ghost, the extension and deletion transfers of Lemmas 3.7 and 3.8 make it an immediate, strictly earlier real attacker win. In either event the theorem's conclusion holds. At an ordinary OR node, discharge the placement role after its deadline check and descend to the exact child.

Step A: an AND node N and a real defender placement d . Exactly one of three cases applies. The cell d is ghost-occupied, it is ghost-empty and searched, or it is ghost-empty and unsearched. Before the cases, fix the following operation for any step on which the ghost cannot copy d .

Filler subroutine. Choose any $d_0 \in S(N)$. Such a cell exists independently of the current real reply by Definitions 4.1 and 4.8. It is empty and legal in the ghost position. If $d_0 \in X$, placing it in the ghost removes d_0 from X . Otherwise the placement adds d_0 to Y . Recompute the canonical differences after the move. Every new Y -cell is a searched ghost filler, and this operation can only shrink X .

Case A1: d is ghost-empty and searched. Both games place at d and the ghost descends to its exact certificate child. The two placements add the same defender stone, so X and Y do not change. Ply, mover, and budget advance identically under Definition 2.5.

Case A2: d is ghost-occupied. The attacker stones are shared and d is empty in the real position. Thus $d \in Y$. The real placement cancels this difference and removes d from Y . Run the filler subroutine in the ghost and descend to the exact child of the selected filler. The two games again consume one defender placement and retain the same mover and budget.

Case A3: d is ghost-empty and unsearched. First suppose d is ghost-legal. If it were protected, then $d \in Z_{\text{dir}}(N)$, and Item (Z2) would put it in $S(N)$. Hence a ghost-legal dismissal is outside the current protected set. It is also outside $Z_{\text{seed}}(N)$, so it satisfies the seed hypothesis of Lemma 4.9 for every protected cell that is not yet ghost-legal.

Now suppose d is ghost-illegal. Its real legality must come from a chain of real-only defender stones. The first protected-occupation lemma says that placing it cannot be the first real-only occupation of a currently protected cell. It is therefore outside the current protected set in this case as well.

The real game places at d . Add d to X and to the monotone history $\hat{X}$. Run the filler subroutine and descend to its child. The ranked seed argument and reachable-obligation nesting preserve $X \cap \text{Prot} = \emptyset$. Occupancy is accounted for by the canonical update, and the radius-8 chain accounts for the legality frontier. These are channels C1 and C3 of Lemma 3.5. The fixed-window conclusion of Lemma 4.11 accounts for channel C2: no real defender window completes before the mapped attacker wins or first enters that window.

A WIN leaf. Let W be a named attacker-alive witness. Every ghost attacker stone in W is shared. Each other cell of W belongs to $E(W, P_L)$ and carries a live witness-empty role, so it is not in X at leaf entry. Since W is attacker-alive in the ghost, it contains no ghost defender stone and hence no Y -cell. The complete real and ghost masks of W , including their empty sets, agree at leaf entry.

The witness has one empty when its count is 5, or two empties when its count is 4 and the attacker budget is 2. By Lemma 3.1, these empties lie within distance 2 of permanent shared attacker stones. They are legal in the real position, remain empty, and can be filled without an intervening defender placement. The real attacker completes W within the current turn and by the leaf's declared resolution.

A LOSS leaf. Apply the same compressed-obligation argument to every $W \in \mathcal{T}$. All attacker stones are shared, every ghost empty of W is protected through leaf entry and is therefore X -free, and attacker-aliveness excludes a Y -cell. Thus the real and ghost masks and empty sets of every named witness agree at leaf entry. The real empty-set family has the checked transversal number

$$\tau(\mathcal{T}) > b.$$

The real defender cannot satisfy `own_win_now` at this leaf. Consider a putative defender witness window. If it has no ever-dismissed fill, the mask inequality (MI), the shared attacker mask, and the equal budget transfer the same immediate-win predicate to the ghost, contrary to the mandatory leaf check. If it has an ever-dismissed fill, the first-dismissal and exposure argument in Lemma 4.11 includes all b placements remaining at the LOSS leaf and precludes completion during the current defender turn.

After the leaf-entry checks, discharge the witness-empty roles. Let the real defender choose any complete nonterminal remainder H of its turn. Since $|H| \leq b < \tau(\mathcal{T})$, this set of placements cannot hit every named empty set. There is some $W_H \in \mathcal{T}$ with

$$E(W_H, P_L) \cap H = \emptyset.$$

The window remains alive for A. It has at most two surviving empties, and Lemma 3.1 places them within distance 2 of permanent shared attacker stones. The attacker fills them during the following turn and completes W_H by leaf- $\text{ply} + b + 2$. No obligation protection is needed after leaf entry; the universal choice of W_H supplies a witness that survived the actual remainder.

Every nonterminal operation descends one edge of the finite certificate. Thus, unless the real attacker has already won strictly earlier, the coupled play reaches an OR-COMPLETION, WIN, or LOSS transfer and resolves by that path's declared ply. The global T is the maximum of these declared resolutions. The compiled strategy wins against every legal real defender play by T . $\square$

6 Computable zones and local reductions

The certificate definitions describe finite verifier checks, but a solver still needs a direct rule for constructing the mandatory searched set. The ranked zone theorem gives that rule at an ordinary AND node. Two narrower reductions apply when the current threats force a fresh equal-position tree, or when a dismissal carries a transition-specific substitution envelope.

6.1 The ranked zone theorem

The four sets from Definition 4.8 account for direct occupancy, legality chains, touched completion windows, and virgin completion windows. Their union is the complete ordinary verifier zone.

Theorem 6.1 (Ranked zone reading). At an ordinary AND node N , define

$$Z(N) = Z_{\text{dir}}(N) \cup Z_{\text{seed}}(N) \cup Z_{\text{touch}}(N) \cup Z_{\text{virgin}}(N).$$

Any independently nonempty searched set $S(N) \supseteq Z(N)$, together with Item (Z4), is sufficient for Theorem 5.1. If $Z(N)$ is empty, the certificate grammar still requires one arbitrary legal searched fallback. The uniform admissible assignment $r_N(y) = B(N)$ replaces Z_{seed} by

$$\text{Legal}(P_N) \cap B_{8(B(N)-1)}(\text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N))).$$

Exact cell ranks can only reduce that uniform obligation band. Exact window exposures likewise reduce the completion guards obtained from the uniform B clock. Larger admissible upper bounds remain sound. Current threat-hitting cells are optional search candidates, not terms of $Z(N)$.

In plain terms. At an ordinary node the verifier takes the union of four finite sets. The solver may search more cells, but it may not omit a cell of that union unless it supplies a valid substitution envelope. An empty union is not an empty AND node: one legal fallback is still needed to advance the ghost play.

Proof. Let $S(N)$ be independently nonempty and contain $Z(N)$. Since it contains $Z_{\text{dir}} \cup Z_{\text{touch}} \cup Z_{\text{virgin}}$, it satisfies Item (Z2). Since it contains Z_{seed} , it satisfies clause (Z5'). The remaining zone-carrying condition is the stated well-formed-legality check Item (Z4). The node therefore satisfies Definition 4.8, and Theorem 5.1 gives pathwise dismissal soundness.

The empty-union case still falls under Definition 4.1, whose AND grammar requires $S(N) \neq \emptyset$ independently of the coverage clauses. For the comparison of clock choices, an exact role rank is bounded by every admissible uniform rank assigned to that role. The ball radius $8(r_N(y) - 1)$ is monotone in the rank, so replacing the exact rank by $B(N)$ can only enlarge the seed band. The recurrences in Definitions 4.2 and 4.5 give $E_N^D(W) \leq B(N)$, and both the touched threshold and virgin radius are monotone in the exposure. Thus exact exposure cannot enlarge the corresponding uniform completion guard. Conversely, replacing any exact label by a larger verifier-checked upper bound preserves every inequality in Lemmas 4.9 and 4.11. No part of this argument uses the current threat-hitting set. $\square$

6.2 The full-clock calculation in the running example

First read the certificate fragment of Fig. 7 with the ordinary D11 full clocks, before replacing its root by a forcing gate. Its eight future attacker and LOSS-witness carriers are

$$\text{Prot}(P_0) = \{(3, 20), (0, 20), (1, 20), (2, 20), \\ (0, 21), (0, 22), (1, 19), (2, 18)\}.$$

Every listed cell is legal at P_0 , and every role has exact root rank one. It follows that

$$Z_{\text{dir}}(P_0) = \text{Prot}(P_0), \quad Z_{\text{seed}}(P_0) = \emptyset.$$

Along the sole certificate branch, for any defender-alive queried window V ,

$$E_{P_0}^D(V) = \begin{cases} 1, & V \cap \{(3, 20), (0, 20)\} \neq \emptyset, \\ 3, & V \cap \{(3, 20), (0, 20)\} = \emptyset. \end{cases}$$

Enumerating the windows incident with the root defender stones leaves exactly four touched qualifiers,

$$V_j = \{(q, 40) : j \leq q \leq j + 5\}, \quad j = -3, -2, -1, 0.$$

Each has defender count three and exposure three. Their empty-cell union is

$$Z_{\text{touch}}(P_0) = \{(q, 40) : q \in \{-3, -2, -1, 3, 4, 5\}\}.$$

No all-empty window has exposure at least six, so $Z_{\text{virgin}}(P_0) = \emptyset$. Thus the numerical ordinary-zone decomposition has sizes 8, 0, 6, 0 in the order direct, seed, touched, virgin. The actual root is instead a forcing gate and searches its exact kernel $K = \{h\}$, as defined in Section 7.

For

$$W = \{(q, 40) : 0 \leq q \leq 5\},$$

the root has

$$\text{cnt}_D(W, P_0) = 3, \quad h \notin W.$$

The full D16 exposure counts the exact defender reply at the root and the two placements reserved by the subsequent budget-2 LOSS leaf:

$$E_{P_0}^D(W) = 1 + 2 = 3. \tag{1}$$

Consequently the conservative ordinary touched calculation reads

$$\text{cnt}_D(W, P_0) + E_{P_0}^D(W) = 3 + 3 = 6.$$

If this full clock is used at an ordinary zone node, all three empties of W belong to its touched term. The gate clock introduced in Definition 7.3 instead has the verified value

$$Q_{P_0}^D(W) = \max\{1, 0 + 2\} = 2,$$

because the exact copied reply h lies outside W . The resulting fixed-window total is $3 + 2 = 5$.

The second query is

$$W' = \{(q, 0) : 5 \leq q \leq 10\}.$$

It contains one root defender stone, at $(8, 0)$, and the copied reply $h = (5, 0)$ lies in it. The branch-coherent gate recurrence therefore gives

$$Q_{P_0}^D(W') = \max\{1, 1[h \in W'] + 2\} = 3.$$

The three units are attained by h followed by the LOSS-remainder placements $(6, 0)$ and $(7, 0)$. The gate proof in Section 7 explains why the first value may be debited and the second may not.

6.3 A short static cover

The ranked zone may sometimes be covered by a position-local set. This is a restricted theorem, not a general replacement for certificate obligations.

Theorem 6.2 (Static-set coverage for short local budgets). Suppose $B(N) \leq 3$, at most three remaining attacker placements are threat-creating, every named witness-empty role lies in an attacker-touched window, and $Z_{\text{seed}}(N) = \emptyset$. Then

$$r_3(P_N) \cup \{\text{empties of attacker-touched alive windows at } P_N\},$$

where $r_3(P_N)$ is the set of legal empties within distance 3 of a stone, covers $Z(N)$. Without all of these hypotheses, the displayed set is a heuristic candidate generator rather than a certified zone.

In plain terms. Under these simultaneous hypotheses, two easy local sets already contain every mandatory zone term. Removing the short-budget, seed, or witness hypothesis can restore remote obligations, so this is not a general replacement for the ranked zone.

Proof sketch. Every exposure is at most $B(N) \leq 3$, so the virgin term is empty. A touched window in the completion term has at least $6 - E_N^D(W) \geq 3$ defender stones, and Lemma 3.1 places each of its empties in $r_3(P_N)$. The first-three containment of Lemma 4.6 covers the stated designated moves, while the witness hypothesis covers leaf roles; the seed term is empty by assumption. See [1]. $\square$

6.4 The extendable-hit kernel

At a fresh equal-position node, the current attacker threats can replace the certificate's future obligations for a short forced region. The remaining reply must not merely hit one threat; it must leave a hitting set that fits in the rest of the defender budget.

For the current attacker-threat empty-set family $\mathcal{F}_N$, write

$$\mathcal{F}_N \setminus d = \{E \in \mathcal{F}_N : d \notin E\}, \quad K_b(N) = \{d \in \text{Legal}(P_N) : \tau(\mathcal{F}_N \setminus d) \leq b - 1\}.$$

Theorem 6.3 (Extendable-hit kernel). A kernel-governed region is entered at a certificate node where the real and ghost positions are equal. It uses the following rule until a typed terminal or an $\text{mhs} > b$ handoff. At every internal AND node in the region, require the explicit defender check $\neg \text{own_win_now}$ and

$$\text{mhs}(P_N) = \tau(\mathcal{F}_N) \leq b,$$

and search exactly $K_b(N)$. No original obligation or core term is required. If $\text{mhs} < b$, then $K_b(N) = \text{Legal}(P_N)$ and there is no pruning. If $\text{mhs} = b$, the kernel weakly refines the current hitting-cell set and can strictly reduce it; equality can occur. A node with $\text{mhs} > b$ ends the kernel region. It must be a valid LOSS leaf or hand off to an existing nonempty sound subtree, below which the core-free claim is not asserted without a new equal-position entry.

In plain terms. The kernel keeps only a first hit that can be extended to a complete defense of the current threat family within the remaining placements. When the family needs fewer hits than the budget, every legal reply is extendable and the rule saves nothing. When the pressure exceeds the budget, the node must use a LOSS contract or return to a separately certified subtree.

Proof sketch. If $\tau < b$, deleting the sets hit by any legal d leaves transversal number at most $b - 1$, while if $\tau = b$ the first member of a minimum transversal is legal by Theorem 3.3 and lies in the nonempty kernel. At the first reply outside the kernel the positions still agree, so the original subtree may be abandoned: for $b = 2$ the residual family gives a budget-1 LOSS, and for $b = 1$ one threat survives for the following budget-2 attacker turn. The explicit immediate-win exclusion bounds every defender-alive window by three stones at $b = 2$ or four at $b = 1$, so the defender cannot complete first. Following a minimum transversal through the searched kernels kills all current threats and leaves the original horizon enough time for the auxiliary two-placement refutation. If $\tau > b$, the kernel is empty, which explains the required handoff. See [1]. $\square$

6.5 Ranked relevance closure

The mandatory closure is the ranked zone itself. A search implementation may add three finite local heuristics without making them verifier premises.

Definition 6.4 (Ranked relevance closure). Define

$$R_{\text{cert}}(\mathcal{C}, N) = Z_{\text{dir}}(N) \cup Z_{\text{seed}}(N) \cup Z_{\text{touch}}(N) \cup Z_{\text{virgin}}(N).$$

The solver may search the finite superset

$$R_{\text{search}}(\mathcal{C}, N) = R_{\text{cert}}(\mathcal{C}, N) \cup \text{hitting}(P_N) \cup \mathcal{A}(P_N) \cup r_3(P_N),$$

where $\text{hitting}(P)$ is the union of the empty sets of current attacker threats, $\mathcal{A}(P)$ is the set of empties in attacker-touched alive windows, and $r_3(P)$ is the legal radius-3 neighborhood of the current stones. The last three terms are heuristics, not hypotheses of Theorem 5.1. The mandatory terms are finite: obligations come from a finite certificate, touched windows from a finite position, and each virgin query examines only windows within its bounded exposure radius of a finite legal candidate.

In plain terms. R_{cert} is the verifier boundary; every cell in it must be covered. R_{search} may be larger to help discovery, but its three added sets are not soundness premises. Omitting a heuristic cell can lose a proof, whereas omitting a mandatory cell can make an accepted proof unsound.

The next theorem turns this computed closure into a verifier condition.

Theorem 6.5 (Ranked-closure coverage). At every ordinary zone-governed AND node, any independently nonempty $S(N) \supseteq R_{\text{cert}}(\mathcal{C}, N)$ satisfies Item (Z2) and clause (Z5'). Dismissal soundness follows from Theorem 5.1 when the D9 grammar, the D14–D16 clock checks, and Item (Z4) also hold. In constructing R_{cert} , the verifier unions roles over all reachable descendants, includes every designated attacker move including OR-COMPLETION, treats WIN and LOSS continuation cells as leaf-entry witness-empty roles, takes the maximum rank over live roles sharing a cell, and forms no band after the $r = 0$ deadline check. Defender-completion windows remain on the E^D clock or the conservative B clock.

In plain terms. Covering R_{cert} is sufficient only together with an independent searched fallback, the certificate grammar, the clock checks, and attacker legality. The statement validates a supplied certificate; it does not promise that a heuristic candidate generator will find one.

Proof sketch. The four terms in R_{cert} are exactly the four sets in Definition 4.8, so coverage supplies Z2 and Z5'. The stated role-union and deadline rules are the hypotheses needed to compute those sets rather than approximations to them. With Z4 and the certificate clocks checked, Theorems 5.1 and 6.1 apply. See [1]. $\square$

6.6 Branch-indexed substitution

The global zone protects the union of every reachable branch. A dismissal may instead select one searched child and verify a smaller envelope over all descendants of that child. The current real and ghost placements are already different, so every selected clock must include that transition.

Definition 6.6 (Transition-inclusive substitution envelope). At an AND node N , a ghost-legal dismissal d may name a searched substitute

$$\varphi_N(d) = s \in S(N)$$

with exact child C_s . The node also declares a nonempty fallback $F(N) \subseteq S(N)$ independently of these annotations. In particular, $S(N)$ is not defined circularly as the replies for which no safe substitute exists. A valid selected envelope has the following data and checks.

- (1) Its transition budget is $\widehat{B}(N, d, s) = 1 + B(C_s)$. It includes the current real/ghost defender ply and every reachable LOSS remainder.

- (2) Its obligation set is the union of the roles at all reachable descendants of C_s , not the roles of one leaf or one selected continuation.
- (3) The real cell d avoids every selected obligation cell and every transition-dangerous completion empty.
- (4) If a selected child obligation y has rank $r_{C_s}(y)$ and $y \notin \text{Legal}(P_N) \cup \text{Stones}(P_N)$, direct occupation $d = y$ is forbidden and the parent seed test is

$$d(d, y) > 8r_{C_s}(y) = 8((1 + r_{C_s}(y)) - 1).$$

- (5) For every defender-alive touched window W with $\text{cnt}_D(W, P_N) \geq 1$, the $\widehat{B}$ test forbids $d \in E(W, P_N)$ whenever $\text{cnt}_D(W, P_N) + 1 + B(C_s) \geq 6$.

For an all-empty window it forbids a seed satisfying $d(d, W) \leq 8(\widehat{B} - 6)$ when $\widehat{B} \geq 6$. The exact-exposure form instead sets

$$\widehat{E}^D(W) = 1 + E_{C_s}^D(W)$$

and uses $\text{cnt}_D(W, P_N) + \widehat{E}^D(W) \geq 6$ for a touched fill and $d(d, W) \leq 8(\widehat{E}^D(W) - 6)$, with $\widehat{E}^D(W) \geq 6$, for an all-empty window. Both touched and virgin clauses are mandatory for the selected clock.

- (6) The fallback $F(N)$, and hence $S(N)$, remains nonempty independently of whether any dismissal has a valid substitute.
- (7) For a ghost-legal A3 dismissal, the ghost plays $\varphi_N(d)$. Case A2 may use any searched filler because it creates no new X -stone. A ghost-illegal A3 dismissal inherits the envelope of the earlier ghost-legal seed that supports its real legality and may use any searched filler in the current reachable subtree. It does not start a child-only clock.
- (8) The envelope protects every reachable LOSS witness-empty role through leaf entry, while $\widehat{B}$ or $\widehat{E}^D$ counts the leaf's b placements for defender completion and defender-own-win exclusion.

The default-child form declares $f(N) \in F(N)$ and uses $C_{f(N)}$ for every ghost-legal dismissal. It must satisfy the same transition-inclusive checks. Replacing a whole-tree role union by a child union without these checks is insufficient.

In plain terms. The real defender may play d while the ghost follows the searched reply s , so the verifier protects the chosen child's entire future against that new mismatch. The leading unit charges the current transition in both the role and window clocks. A child-only calculation would miss a legality link or a defender fill created on this ply.

The envelope is useful only if it preserves the main dismissal conclusion.

Theorem 6.7 (Branch-indexed substitution soundness). The conclusion of Theorem 5.1 remains valid if an AND node's global Definition 4.8 dismissal tests are replaced, dismissal by dismissal, by a valid Definition 6.6 envelope. The same holds for the default-child form.

In plain terms. A valid envelope permits branch-specific substitution, not an arbitrary reply that happens to hit the same threat. The descendant union and transition-inclusive clocks preserve all three causal channels. If any check fails, the node is not certified by this theorem.

Proof sketch. The real reply and ghost substitute consume the same defender ply, while the transition cell and the leading 1 account for occupancy, legality-frontier, and mask differences at that step. Every later ghost node lies among the declared descendants of C_s ; nested selected envelopes preserve avoidance for earlier X -stones, and a ghost-illegal chain inherits the envelope of its first ghost-legal seed. The touched and virgin tests give the fixed-window argument through every reachable LOSS remainder. Z4, leaf-mask transfer, and the terminal cases of Theorem 5.1 are unchanged. See [1]. $\square$

The leading 1 is needed in both hazard channels. For occupancy, a legal real reply at distance 8 from a future ghost-illegal obligation can legalize that obligation for the remaining defender placement; a child rank of one would give radius zero without the transition and radius eight with it. For completion, two defender stones in a window, a real transition into that window, and three later fills total six even though the child-only count is $2 + 3 = 5$.

The ordinary completion zone also derives the internal immediate-win diagnostic.

Lemma 6.8 (Internal own-win diagnostic). Under the completion-zone requirement of Definition 4.8 and the ban in Definition 4.1 on defender-terminal edges, an internal AND node cannot contain a defender-alive count-5 window, or a count-4 window when the defender budget is 2.

Proof sketch. A count-5 touched window has exposure at least one, so its legal last empty is searched and its exact child would be defender-terminal. At budget 2, a count-4 touched window has exposure at least two, so both empties are searched; either first fill reaches a budget-1 count-5 child. Such a child cannot be a LOSS leaf because its mandatory immediate-win check fails, and an internal child would again search the terminal last empty. See [1]. $\square$

The check remains mandatory at LOSS leaves and in the kernel theorem, whose governed region has no ordinary completion guard.

6.7 Finite certificate DAGs

Identical certificate subtrees may be shared, provided every route into a shared node agrees on its exact state and clock.

Definition 6.9 (Finite acyclic certificate DAG). A D9 certificate may be a finite rooted acyclic graph if every shared node has one exact D9 label: the same position, mover and budget, designated OR action or AND searched-successor map, leaf witness data, and one consistent path clock. Obligations at a DAG node are unions over all reachable descendants. Local budgets, ranks, and exposures satisfy their inequalities on every outgoing edge. Coupling histories X , Y , and $\hat{X}$ remain path-local.

In plain terms. A DAG may share a certificate state only when every incoming route agrees on the state, action, witnesses, and clocks. The stored node is shared, but the real-ghost difference history remains attached to a path. Merging inconsistent labels would no longer describe one certificate node.

Soundness follows by expanding the shared representation back into a tree.

Theorem 6.10 (DAG unfolding). Every Definition 6.9 DAG satisfying the ordinary Definition 4.8 rules or the substitution rules of Definition 6.6 has the corresponding soundness conclusion of Theorem 5.1.

In plain terms. The DAG saves storage but proves no new play. Unfolding makes one copy of a shared node for each incoming path and leaves every verified label unchanged. A cycle or inconsistent clock would prevent this finite tree reduction.

Proof sketch. A finite acyclic graph has finitely many root-to-node paths. Unfold one copy of each shared node along each path; the exact labels and consistent clocks give a finite D9 tree with unchanged transitions, terminal data, clock inequalities, and descendant-role unions. Apply Theorem 5.1 or Theorem 6.7; the unfolded coupling histories are precisely the required path-local histories. See [1]. $\square$

At verification time, an ordinary node must establish R_{cert} coverage and Z4, or validate every substitution annotation, while retaining an independent fallback. A failed check returns UNKNOWN. Heuristic search trimming can affect completeness, but a certificate that passes these checks remains sound.

6.8 Local domination patterns

The zone theorems compare a real reply with a ghost certificate path. Four additional local rules compare two defender replies directly. Their proofs use causal strategy transfer and check the same occupancy, window-mask, and legality-frontier channels.

For this subsection, let

$$\Lambda(P) = \bigcup_{s \in \text{Stones}(P)} B_8(s)$$

be the supported cells, and let $\Omega(x)$ be the 18 windows containing x . For legal defender replies a, b at the same position, write $b \preceq_n a$ when, for every defender continuation strategy after b , there is one after a such that, for every attacker challenge after a , there is an attacker continuation after b satisfying

$$u_D(\text{Out}_n(P + a, \sigma_a, \tau_a)) \geq u_D(\text{Out}_n(P + b, \sigma_b, \tau_b)),$$

where u_D assigns 1 to a defender win, 0 to no terminal event through the horizon, and -1 to an attacker win. Thus an attacker win against the searched reply a transfers to the omitted reply b .

A dead empty cell supplies no new radius-8 legality frontier.

Lemma 6.11 (A dead empty has no legality frontier). *Let b be empty and dead in a nonterminal post-opening position P , meaning that every window in $\Omega(b)$ is dead. Then*

$$B_8(b) \subseteq \Lambda(P).$$

In particular, b is legal, and placing at b adds no unsupported cell to the legal store.

Proof sketch. For each directed axial unit vector u , the endpoint window $\{b, b + u, \dots, b + 5u\}$ is dead and contains an old stone $s_u = b + k_u u$ with $1 \leq k_u \leq 5$. A cell $z \in B_8(b)$ lies in a 60° sector between adjacent directions; choosing the stone on either boundary ray gives $d(z, s_u) \leq 8$. Hence every point of $B_8(b)$ was already supported. See [1]. $\square$

Deadness of the omitted cell does not make an arbitrary searched substitute safe. The substitute must also be frontier-inert unless it wins immediately.

Proposition 6.12 (Frontier-certified dead-cell dismissal). *Let P be a nonterminal post-opening position with D to place, and let $a \neq b$ be legal empty cells. Assume every window in $\Omega(b)$ is already dead. Unless $P + a$ is an immediate defender win, also assume*

$$\Lambda(P) \cup B_8(a) = \Lambda(P) \cup B_8(b).$$

Then $b \preceq_n a$ for every finite n . By Lemma 6.11, the displayed support equality is equivalent here to $B_8(a) \subseteq \Lambda(P)$.

Proof sketch. Compare $P + a$ and $P + b$ by transposing the two special coordinates until the counterpart is used. The support equality makes the transposition legal at every paired prefix. All windows through b are permanently dead; the other changed windows either agree or change ownership in the defender-favouring direction. Per-placement terminal checks stop the transfer at the first win, and an immediate defender win after a is already maximal. This gives an all-depth alternating simulation and hence the stated domination. See [1]. $\square$

Two hits of one count-4 threat are interchangeable only when their other incident windows and their successor supports are made inert.

Proposition 6.13 (Dead-spoke interchangeable hits). *Let P be a nonterminal post-opening position with D to place. Let W be an attacker-alive window with exactly four attacker stones and exactly two empty cells $x \neq y$. Both cells are legal. Assume every window in*

$$(\Omega(x) \cup \Omega(y)) \setminus \{W\}$$

is already dead in P , and assume

$$\Lambda(P) \cup B_8(x) = \Lambda(P) \cup B_8(y).$$

Then the two replies are mutually outcome-dominating at every finite horizon:

$$x \preceq_n y, \quad y \preceq_n x.$$

A solver may retain one canonical member of $\{x, y\}$ at that defender node.

Proof sketch. Either defender hit kills W , and the dead-spoke premise then makes every window through either special cell permanently dead. Transposing x and y preserves occupancy choices, the support equality preserves legal moves in both directions, and all masks outside the dead union agree. This is an outcome-preserving bisimulation at every depth. Equality of window count profiles alone would not preserve labelled empties or their intersection hypergraph. See [1]. $\square$

The two placements of one turn can be stored as an unordered pair only if both orders were legal before the first placement and neither singleton prefix wins.

Proposition 6.14 (Nonwinning-prefix turn commutation). *Let P be a nonterminal FirstStone position for either player X . Let $p \neq q$ both be legal in P , and suppose neither transition $P + p$ nor $P + q$ is terminal. Then both two-placement traces $p; q$ and $q; p$ are legal and have the same qualitative rule outcome after the second placement. If nonterminal, they have identical occupancy, labelled window masks, legal support, player to move, and FirstStone phase. If terminal, both terminate on the second placement with winner X and the same absolute placement count.*

Proof sketch. Initial legality makes either second cell remain legal, and the final support $\Lambda(P) \cup B_8(p) \cup B_8(q)$ is independent of order. Adding the two owner bits to every window mask commutes, including on a window containing both cells. The two singleton-nonterminal premises exclude a first-placement win, so any joint completion occurs on the second placement in both orders. The live engine states may retain different order-sensitive history metadata; the proposition equates rule outcomes, not serialized state fields. See [1]. $\square$

7 The forced-hit gate calculus

The ordinary zone charges every future defender placement before a role or window deadline. A tight forcing line contains replies that the defender must copy into the certificate or abandon the line. An exact copied reply does not create a real-only frontier stone. This permits a narrower hazard clock. The escape branch still needs the full remaining defender turn. The construction and coordinate checks in this section are recorded in the companion proof and forced-hit report [1].

7.1 Why pressure alone cannot be debited

The first example isolates the terminal race. The current threats appear to consume both defender placements, but the defender can ignore them and win before the attacker receives a turn.

Example (Reachable disjoint-hit race). Let

$$\begin{aligned} W &= \{(0, r) : 0 \leq r \leq 5\}, \\ T_1 &= \{(q, -4) : 0 \leq q \leq 5\}, \quad a = (5, -4), \\ T_2 &= \{(q, 8) : -5 \leq q \leq 0\}, \quad b = (-5, 8). \end{aligned}$$

The following legal history begins with a defender opening at the origin. Each row after the opening is one two-placement turn.

Plies	Mover	Placements
0	D	(0, 0)
1–2	A	(0, -4), (1, -4)
3–4	D	(0, 1), (-1, -4)
5–6	A	(2, -4), (3, -4)
7–8	D	(0, 2), (0, -8)
9–10	A	(4, -4), (-4, 8)
11–12	D	(0, 3), (1, 8)
13–14	A	(-3, 8), (-2, 8)
15–16	D	(-8, 0), (8, -8)
17–18	A	(-1, 8), (0, 8)

Every placement is within distance eight of an earlier stone. No displayed prefix completes a window. The resulting node has defender budget $b = 2$ and complete attacker threat-empty family

$$\mathcal{F} = \{\{a\}, \{a, (6, -4)\}, \{b\}, \{b, (-6, 8)\}\}, \quad \tau(\mathcal{F}) = 2.$$

The singleton sets force every transversal to contain a and b , so the extendable-hit kernel is exactly $\{a, b\}$. It is disjoint from W . The defender may ignore both threats and play

$$u = (0, 4), \quad v = (0, 5).$$

Both cells are legal at turn start. The first raises $\text{cnt}_D(W)$ to five, and the second completes W . Termination occurs before the attacker can use either surviving threat.

The same construction can be translated or reflected, and remote legal fillers can be added. It is not a boundary artifact. A sound debit must distinguish a copied hit from a refusal and must charge the refusal branch for the complete remaining turn. Threat masks also have to agree between the real and ghost positions at the gate. Otherwise a real-only defender stone can kill one named threat and reduce the actual pressure.

Substitution raises a separate obstruction. A real hit and a different ghost hit can answer the same threats while creating nonempty X and Y sets. The real stone may then support a radius-eight relay that the ghost stone does not support. Such a hit remains an ordinary transition for the frontier clock. Only an exact copied placement receives the gate debit.

7.2 Protected exact-copy gates

A gate names the current pressure and protects its masks through an entry checkpoint. It then searches an exact residual-transversal kernel rather than an ordinary four-part zone.

Definition 7.1 (Protected exact-copy forcing gate). At an internal AND node Q with defender budget $b \in \{1, 2\}$, a *forcing gate* names a finite family $\mathcal{H}_Q$ of current attacker threat windows. Put

$$\mathcal{F}_Q = \{E(U, P_Q) : U \in \mathcal{H}_Q\}.$$

The verifier checks

$$\tau(\mathcal{F}_Q) = b, \quad \neg \text{own_win_now}(P_Q). \quad (\text{FG1--FG2})$$

For a cell d , define

$$\mathcal{F}_Q \setminus d = \{E \in \mathcal{F}_Q : d \notin E\}, \quad K(Q) = \{d \in \text{Legal}(P_Q) : \tau(\mathcal{F}_Q \setminus d) \leq b - 1\}.$$

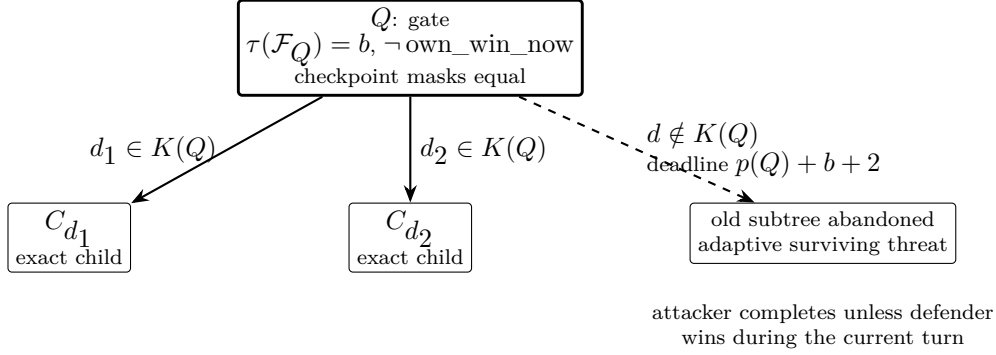


Figure 9: Forcing-gate grammar. Kernel edges are exact copies. The dashed escape is an adaptive terminal contract, not a substituted certificate child.

For every $U \in \mathcal{H}_Q$ and $e \in E(U, P_Q)$, add a checkpoint role (Q, U, e) to the obligation grammar. Its deadline is the gate-entry mask check immediately before the defender reply. It belongs to every strict ancestor's reachable obligation union and remains protected through that check. Write $\text{Prot}^-(Q)$ for the incoming protected set. After the verifier checks the named ghost masks, the checkpoint roles are discharged and the remaining set is $\text{Prot}^+(Q)$.

The searched set at the gate is exactly the finite nonempty set $K(Q)$. Each $d \in K(Q)$ has its exact successor C_d , and a continuing real reply is copied on that edge. No substitution envelope is permitted on a kernel reply. A real reply outside $K(Q)$ abandons the original subtree and uses the adaptive escape contract of Lemma 7.5.

Let $p(Q)$ be the absolute index of the last placement before entry to Q . The escape deadline is

$$p(Q) + b + 2.$$

The certificate horizon is the maximum of its ordinary declared resolutions and all reachable escape deadlines.

In plain terms. The checkpoint makes every named threat the same in the real and ghost positions before the gate is used. A reply inside the kernel is copied at the same cell. A reply outside it discards the old plan and gives the attacker a surviving current threat, unless the defender wins first during the remainder of the turn.

The kernel is nonempty. If T is a size- b minimum transversal, each $d \in T$ leaves $T \setminus \{d\}$ as a transversal of $\mathcal{F}_Q \setminus d$, and Theorem 3.3 makes d legal. It is finite because a kernel cell must belong to some named empty set. At $b = 1$, one threat is enough. At $b = 2$, a subfamily of at most three threats can retain transversal number two.

The role clock counts opportunities to create a real-only defender stone. An exact gate edge creates none.

Definition 7.2 (Forced-hit-debited role ranks). For a role ρ live at N , define $f_N(\rho)$ in reverse topological order by

$$f_N(\rho) = \begin{cases} 0, & \rho \text{ is at its deadline or no longer reachable,} \\ f_C(\rho), & N \text{ is an ordinary OR and } \rho \text{ remains live,} \\ 1 + \max_C f_C(\rho), & N \text{ is an ordinary AND node,} \\ \max_{d \in K(N)} f_{C_d}(\rho), & N \text{ is a forcing gate.} \end{cases}$$

A child where ρ is not reachable contributes zero. For a cell carrying several live roles, set

$$f_N(y) = \max\{f_N(\rho) : \rho \text{ is live at } N \text{ and carried by } y\}.$$

An ordinary AND edge costs one. An exact gate edge costs zero because its defender stone is shared. An off-kernel reply abandons every old role.

In plain terms. The value f is not the number of defender plies before resolution. It counts only placements that can extend a real-only legality chain toward a protected cell. Full path length continues to use the local budget B .

A copied gate hit can still harm a defender window when the hit lies inside that window. The exposure recurrence therefore keeps a window-specific indicator and a full escape floor.

Definition 7.3 (Forced-hit-debited window exposure). For a forcing gate, write

$$G_N(W) = \max_{d \in K(N)} (\# [d \in W] + Q_{C_d}^D(W)).$$

For every window W , define $Q_N^D(W)$ by

$$Q_N^D(W) = \begin{cases} 0, & N \text{ is WIN or OR-COMPLETION,} \\ b, & N \text{ is a LOSS leaf with defender budget } b, \\ 0, & N \text{ is an OR whose attacker placement enters } W, \\ Q_C^D(W), & N \text{ is any other ordinary OR,} \\ 1 + \max_C Q_C^D(W), & N \text{ is an ordinary AND,} \\ \max\{b, G_N(W)\}, & N \text{ is a forcing gate.} \end{cases}$$

If W is already not alive for the defender, set $Q_N^D(W) = 0$. The maximum at a gate is taken child by child. Replacing it by independent maxima is conservative but can overcount by one.

On a continuing path, let F count every ordinary defender opportunity, every remaining defender placement at a LOSS leaf, and all b placements of the first off-kernel escape turn. Let H_W count exact copied gate replies whose cells lie in W . The recurrence is the branch-coherent maximum of $F + H_W$. An ordinary reply is counted in F even when it hits a threat.

In plain terms. A copied hit outside W costs zero to that window. A copied hit inside W costs one. Refusing a gate costs the full remaining turn, and a LOSS leaf also costs its full remaining budget. These cases prevent the race from the opening example.

The ordinary four-part zone now uses the debited hazard clocks. A gate itself does not use these four terms.

Definition 7.4 (Ordinary debited zone). At an ordinary internal AND node, define

$$\begin{aligned} Z_{\text{dir}}^{FH}(N) &= \text{Prot}(N) \cap \text{Legal}(P_N), \\ Z_{\text{seed}}^{FH}(N) &= \bigcup_{\substack{y \in \text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N)) \\ f_N(y) \geq 1}} \left(\text{Legal}(P_N) \cap B_{8(f_N(y)-1)}(\{y\}) \right), \\ Z_{\text{touch}}^{FH}(N) &= \bigcup \left\{ E(W, P_N) : \begin{array}{l} W \text{ is defender-alive, } \text{cnt}_D(W, P_N) \geq 1, \\ \text{cnt}_D(W, P_N) + Q_N^D(W) \geq 6 \end{array} \right\}, \\ Z_{\text{virgin}}^{FH}(N) &= \left\{ c \in \text{Legal}(P_N) : \begin{array}{l} \text{some all-empty } W \text{ has } Q_N^D(W) \geq 6, \\ d(c, W) \leq 8(Q_N^D(W) - 6) \end{array} \right\}. \end{aligned}$$

The node searches an independently nonempty superset of the union of these four sets. Its obligation union includes every reachable checkpoint role, and clause (Z4) is unchanged. A forcing gate performs its entry checks and searches exactly $K(Q)$.

For a fixed augmented certificate,

$$f_N(\rho) \leq r_N(\rho), \quad Q_N^D(W) \leq E_N^D(W).$$

The full B , r , and E^D clocks retain their ordinary AND charges at a gate. Checkpoint roles can enlarge Prot at strict ancestors, so no claim is made that the total searched-set size decreases in every certificate.

In plain terms. The debit can shrink a ranked seed band or a completion guard for a fixed augmented certificate. It does not make the gate an ordinary zone node, and it does not discount a general substitution. Added checkpoint obligations can offset the smaller hazard radii.

7.3 Transfer at a gate

The entry checkpoint makes the named threat masks identical. The residual transversal inequality then covers every reply outside the kernel.

Lemma 7.5 (Protected gate dichotomy). *On entry to a Definition 7.1 gate, every named threat window has identical real and ghost masks. For every real legal defender reply d , exactly one of the following holds.*

- (1) *If $d \in K(Q)$, then d is shared-empty and ghost-legal. Both games place at d , take the exact child, and leave X and Y unchanged.*
- (2) *If $d \notin K(Q)$ and the defender does not win during the remaining current turn, the attacker completes a named surviving threat in at most two placements of the following turn, by $p(Q) + b + 2$.*

Proof. Fix $U \in \mathcal{H}_Q$. Its attacker stones are shared. Every ghost empty $e \in E(U, P_Q)$ carries a checkpoint role, so $e \notin X$ at the entry check. Since U is attacker-alive in the ghost position, it contains no ghost defender stone and hence no Y -cell. The complete real and ghost masks of U agree.

If $d \in K(Q)$, then d belongs to a named empty set. It is shared-empty and is legal by Theorem 3.3. The gate supplies its exact child. This is the first case.

Suppose $d \notin K(Q)$. By definition,

$$\tau(\mathcal{F}_Q \setminus d) > b - 1.$$

Let H be the set of at most $b - 1$ later defender placements in the current turn. It cannot meet every member of $\mathcal{F}_Q \setminus d$. Some named window therefore avoids both d and H . It remains attacker-alive and retains its one or two initial empties. By Lemma 3.1, those empties lie within distance two of permanent shared attacker stones. They are legal, and the attacker completes the window in the next turn. The last possible attacker placement has index $p(Q) + b + 2$. The game rule allows the defender to terminate first during the current turn, which is the alternative retained in the statement. Membership in $K(Q)$ makes the two cases disjoint. $\square$

The weighted clocks bound the two hazards needed by the coupling. They also remain below the corresponding full clocks.

Lemma 7.6 (Weighted hazard bounds). *For every certificate using Definitions 7.1 to 7.4:*

- (1) *on a continuing mapped path, the number of ordinary real-only frontier opportunities before a live role's deadline is at most $f_N(\rho)$;*
- (2) *for a fixed W , before the certificate attacker wins or enters W , or before an off-kernel escape resolves, the weighted count of ordinary defender edges, copied gate edges in W , and all placements in the first LOSS or escape remainder is at most $Q_N^D(W)$;*
- (3) *$f_N(\rho) \leq r_N(\rho)$ and $Q_N^D(W) \leq E_N^D(W)$;*
- (4) *at a gate, $B(Q) \geq b$, so the full local budget covers every escape remainder.*

Proof sketch. Reverse induction follows the displayed recurrences. An ordinary AND edge can add one X -stone and is charged once; a copied gate edge changes neither X nor Y and is charged to W exactly when its cell lies in W . LOSS and escape remainders retain the full base b . Comparing each clause with the ordinary rank and exposure recurrences gives the two inequalities, while the nonempty kernel and the $b = 2$ successor phase give $B(Q) \geq b$. See the companion proof for the child-by-child induction [1]. $\square$

The first-bad-event argument combines the frontier and completion channels.

Lemma 7.7 (Debited first-bad-event lemma). *Under Definitions 7.1 to 7.4:*

- (1) *while the real play remains mapped to the certificate, no defender placement creates a real-only stone in the current protected set; old roles are abandoned when an off-kernel escape starts;*
- (2) *before the mapped certificate attacker or a gate-escape attacker resolves, no real defender play completes a window.*

Proof sketch. For protected occupation, trace a first bad cell back to the last ghost-legal dismissed seed. Copied gate stones are shared and cannot be internal links of a real-only chain, so every link is charged by f and the seed lay in Z_{seed}^{FH} . For a first defender completion, separate no-dismissal, touched, and virgin cases. The mask identity handles no-dismissal lines; Q^D charges every ordinary fill, every copied hit in the fixed window, and each full LOSS or escape remainder. The touched count or virgin distance inequality then places the first dismissed fill or its seed in the applicable zone. The complete first-event analysis is in the companion proof [1]. $\square$

The gate theorem replaces the two ordinary hazard clocks while retaining the rest of the certificate grammar.

Theorem 7.8 (Exact-copy $F + H_W$ soundness). *Let a finite certificate tree satisfying Definition 4.1, or a finite certificate DAG satisfying Definition 6.9, use Definition 7.4 at every ordinary AND node and Definition 7.1 at every forcing gate. Include every checkpoint role in the reachable obligation unions, retain clause (Z4), retain the full local budget B of Definition 4.2, and include every gate escape deadline in the global horizon. Then the compiled attacker wins against every real defender play by that horizon. The debited f and Q^D values are sound replacements for r and E^D in the ordinary debited seed, touched, and virgin terms.*

Proof. At an ordinary node, run the A1–A3 coupling from Theorem 5.1. Use Definition 7.4 and Lemma 7.7 in place of the ordinary protected-occupation and completion lemmas.

At a gate, check the incoming masks and discharge the checkpoint roles. Lemma 7.5 transfers the named threat family. A real reply in $K(Q)$ is shared-empty, is copied on its exact child, and leaves X and Y unchanged. A reply outside $K(Q)$ abandons the mapped subtree. The second part of Lemma 7.7 excludes a defender completion during the remaining turn. The second case of Lemma 7.5 then supplies an adaptive surviving threat that the attacker completes by the escape deadline.

WIN, LOSS, OR-COMPLETION, and ordinary OR transfers are those of Theorem 5.1. A LOSS remainder remains sound because Q^D charges all b placements and the obligation grammar protects every witness empty through leaf entry. A finite tree reaches a typed terminal or its first escape. A finite DAG can be unfolded as in Theorem 6.10; the gate labels, checkpoint roles, and branch-coherent maxima are preserved. Hence every real play resolves by the augmented horizon. $\square$

In plain terms. The theorem permits a discount only after the gate masks have been protected and the real reply is copied at the same cell. Every refusal is charged as an escape, and every ordinary or substituted reply keeps its full transition cost. These conditions preserve the main coupling while making the two hazard clocks target-specific.

An ordinary substitution envelope can coexist with gates, but its transition continues to use full clocks.

Theorem 7.9 (Substitution-envelope compatibility). *The conclusion of Theorem 7.8 remains valid when an ordinary node's global debited-zone tests are replaced by a valid Definition 6.6 envelope, provided the selected reachable-role union includes every future checkpoint role and the envelope retains the transition-inclusive $\hat{B}$, role ranks, and $\hat{E}$ tests of Definition 6.6. No debited substitution envelope is asserted.*

In plain terms. An ordinary substitution still creates a real-ghost mismatch on its current ply, so it receives no gate discount. Its full transition clocks must also protect every later gate checkpoint. Omitting either condition would combine two individually sound reductions without covering their interaction.

Proof sketch. The real reply and ghost substitute create the ordinary X, Y transition, so the current unit remains mandatory. The full envelope covers all three causal channels and every later checkpoint role. Subsequent debited ordinary steps satisfy $f \leq r$ and $Q^D \leq E^D$, while a gate retains the full envelope inequalities and charges any escape floor. The nested-envelope argument of Theorem 6.7 then applies [1]. $\square$

7.4 The running certificate fires its gate

The position introduced in Section 4 has defender budget one, a singleton threat with empty cell $h = (5, 0)$, and kernel $K = \{h\}$. The gate copies h . The attacker then plays $(3, 20)$ and $(0, 20)$ and reaches a defender-budget-two LOSS leaf whose three witness-empty pairs are disjoint. Their transversal number is three.

For the defender-alive window

$$W = \{(q, 40) : 0 \leq q \leq 5\},$$

the root count is three and $h \notin W$. The full exposure is

$$E_{\text{root}}^D(W) = 1 + 2 = 3,$$

so its touched test reads $3 + 3 = 6$. The gate recurrence instead gives

$$Q_{\text{root}}^D(W) = \max\{1, 0 + 2\} = 2,$$

and the debited test reads $3 + 2 = 5$. The value two is attained when the LOSS remainder places at $(3, 40)$ and $(4, 40)$. The window reaches count five, one of the three LOSS witnesses survives, and the attacker completes that witness. The one-unit debit and the LOSS base two are both used on a legal line.

For

$$W' = \{(q, 0) : 5 \leq q \leq 10\},$$

the copied hit lies in the queried window. The later LOSS placements $(6, 0)$ and $(7, 0)$ give

$$Q_{\text{root}}^D(W') = \max\{1, 1 + 2\} = 3.$$

All three placements lie in W' and the count does not reach six. Omitting the indicator for a copied hit inside W' would give two and would undercount this line.

7.5 What remains open

The exact-copy theorem is proved, but it resolves only the two named hazards at protected tight gates. The broader debit objective remains partial.

The limit map in Section 12 records this result as *partially resolved*. That status concerns the intended pruning extension, not the validity of Theorem 7.8.

Open extension	Present boundary
Target-independent budget	No scalar $B_{FH} < B$ is proved. Path length, LOSS remainders, substitution envelopes, and escape remainders still use the full B .
Substituted forced hits	No D17 substitute receives zero cost. A different real and ghost hit can create a real-only frontier corridor.
Net searched-set shrinkage	The f and Q^D terms do not grow for a fixed augmented certificate, but checkpoint roles can enlarge ancestor obligation sets.
Slack pressure	A family with $\tau < b$ does not force the next reply; its kernel is the whole legal set. No additional debit is proved for that turn.
Seed-radius sharpness	Exact sharpness of every f -based reduction remains open. Ordinary paths retain the relative boundary $8(r-1)$.

Table 2: Open extensions of the forced-hit gate calculus.

8 Sparse LOSS witnesses

A LOSS leaf names current attacker threats whose empty sets cannot all be hit within the defender’s remaining budget. Naming the complete threat family is unnecessary. The special geometry of six-cell Hexo windows gives smaller witness families than a general rank-two hypergraph.

Lemma 8.1 (Hexo-sparse LOSS witnesses). *A Definition 4.1 LOSS witness can be chosen with at most three windows when $b = 1$ and at most five windows when $b = 2$. Both bounds are sharp among Hexo threat-window families.*

Proof. Every threat empty set has size one or two.

Suppose first that $b = 1$. If the family contains a singleton $\{a\}$, choose that set and one member missing a . Such a member exists because the full family has transversal number greater than one. If there is no singleton, start with a pair $\{a, b\}$. Choose one member missing a and one member missing b . No single cell hits the selected family. At most three members are used.

Now let $b = 2$. If the family contains a singleton $\{a\}$, let $\mathcal{H}$ be the subfamily of members missing a . If $\tau(\mathcal{H}) \leq 1$, then a together with a one-point transversal of $\mathcal{H}$ would hit the full family. Hence $\tau(\mathcal{H}) > 1$. The $b = 1$ argument selects at most three members of $\mathcal{H}$; adjoining $\{a\}$ uses at most four, and therefore at most five.

It remains to consider a family of pairs. Choose an inclusion-minimal subfamily $\mathcal{G}$ with $\tau(\mathcal{G}) > 2$. A general rank-two selection uses at most six members. To see this, take a maximal disjoint subfamily. Three disjoint pairs already have transversal number three. One pair cannot be maximal, because its two endpoints would then hit every member. In the remaining case take two disjoint members

$$E_1 = \{a, b\}, \quad E_2 = \{c, d\}.$$

Their four two-point transversals are $\{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}$. For each cross-pair, choose one member of $\mathcal{G}$ that it misses. Together with E_1, E_2 , these choices use at most six members and exclude every two-point transversal.

Assume equality six is necessary. The four missing members must be distinct, and each must miss only its assigned cross-pair. A two-set missing $\{a, c\}$ while meeting the other three cross-pairs must be $\{b, d\}$. Using only one of b, d would miss another cross-pair, while using two outside points would be disjoint from both E_1 and E_2 . Cycling this argument gives $\{b, c\}$, $\{a, d\}$, and $\{a, c\}$. Thus a six-member minimal obstruction is exactly the six edges of K_4 .

Such a K_4 cannot be a Hexo threat-empty family. Each empty pair lies on one grid axis by Fact 2.3. To classify four pairwise axis-collinear cells, translate one of them to the origin. If the other three use the three different incident axes, write them as

$$(u, 0), \quad (0, v), \quad (w, -w),$$

with $u, v, w \neq 0$. Pairwise axis alignment forces $u = v = w = -v$, a contradiction. If two of the other cells use the same axis, then three cells already lie on one common axis line. An off-line fourth cell has only two nonparallel axis lines through it that meet this common line, so it cannot align with all three distinct cells. Hence all four lie on one common axis line. Order them $p_1 < p_2 < p_3 < p_4$. Any consecutive six-cell window containing p_1 and p_3 also contains the intervening empty p_2 . Its empty set cannot be the K_4 edge $\{p_1, p_3\}$. This is a contradiction, so $|\mathcal{G}| \leq 5$.

The two constructions below attain three and five, respectively. They show sharpness among Hexo threat-window families. $\square$

In plain terms. A verifier never needs a large collection of LOSS windows. Three named windows suffice with one defender placement left, and five suffice with two. The five-window bound uses Hexo line geometry; a general family of pairs can need the six edges of K_4 .

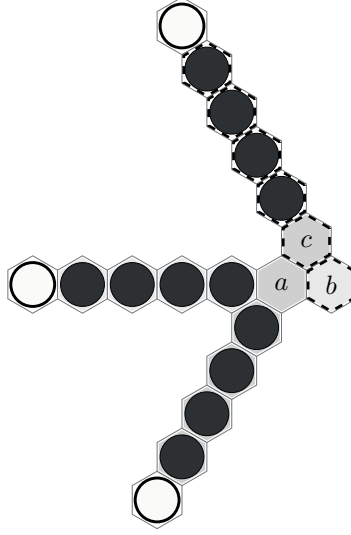


Figure 10: The triangle witness for $b = 1$. The three labelled empty cells give the three pair edges; the open disks are the blockers that remove shifted threats.

8.1 The sharp three-window configuration

Let

$$a = (4, 0), \quad b = (5, 0), \quad c = (4, 1),$$

and define

$$\begin{aligned} W_{ab} &= \{(i, 0) : 0 \leq i \leq 5\}, \\ W_{ac} &= \{(4, i) : -4 \leq i \leq 1\}, \\ W_{bc} &= \{(5 - i, i) : 0 \leq i \leq 5\}. \end{aligned}$$

Place attacker stones on $(W_{ab} \cup W_{ac} \cup W_{bc}) \setminus \{a, b, c\}$ and defender blockers at

$$(-1, 0), \quad (4, -5), \quad (-1, 6).$$

The origin is an attacker stone. Exhaustive window enumeration gives exactly three attacker threats, with empty sets

$$\{a, b\}, \quad \{a, c\}, \quad \{b, c\}.$$

Their transversal number is two. Every subfamily of at most two has a common point, so all three windows are required at a $b = 1$ LOSS leaf. After any one defender placement, one edge remains untouched and the attacker fills its two cells.

8.2 The sharp five-window configuration

Let

$$v_0 = (4, 0), \quad v_1 = (5, 0), \quad v_2 = (6, 0), \quad v_3 = (4, 2), \quad v_4 = (4, 1).$$

Use the five windows

$$\begin{aligned} W_{01} &= \{(i, 0) : 0 \leq i \leq 5\}, & W_{12} &= \{(5 + i, 0) : 0 \leq i \leq 5\}, \\ W_{23} &= \{(6 - i, i) : 0 \leq i \leq 5\}, & W_{34} &= \{(4, 1 + i) : 0 \leq i \leq 5\}, \\ W_{40} &= \{(4, -4 + i) : 0 \leq i \leq 5\}. \end{aligned}$$

Place attacker stones on their union minus the five vertices, and place defender blockers at

$$(-1, 0), \quad (4, -5), \quad (11, 0), \quad (4, 7), \quad (0, 6).$$

The complete attacker-threat family is exactly

$$\{v_0, v_1\}, \{v_1, v_2\}, \{v_2, v_3\}, \{v_3, v_4\}, \{v_4, v_0\}.$$

It is the edge set of C_5 . Its vertex-cover number is three. Removing any edge leaves a four-edge path with vertex-cover number two. Hence no four windows have transversal number greater than two, while all five do. The finite checker enumerates twenty attacker stones, the five blockers, every incident window, the absence of a completion and of a defender immediate win, and exactly these five threats.

The caps are relative to the named-window LOSS format. A smaller cap would reject the displayed valid leaves. The constructions do not rule out a different terminal representation with other witness data.

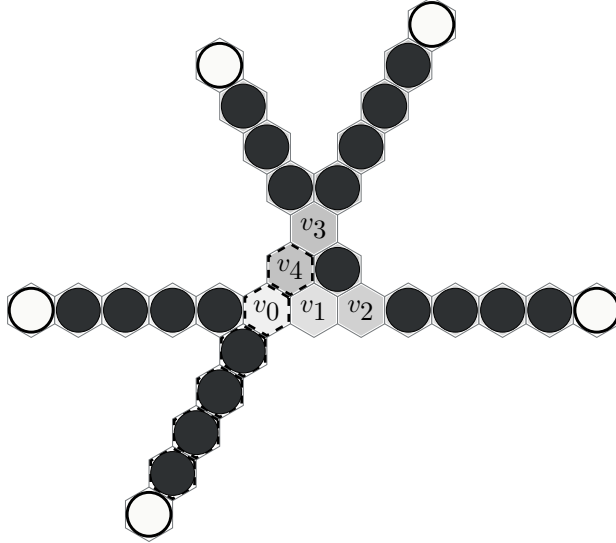


Figure 11: The five-cycle witness for $b = 2$. Each shaded or outlined window has one adjacent pair of labelled empties.

9 Pairing on the hex lattice

A pairing strategy fixes disjoint pairs of cells before play. Whenever the attacker occupies one member of a pair, the defender takes the other. Such a strategy blocks a target window when that window contains both members of at least one pair. The method would reduce a reply choice to one cell, but the density required by the three hex-lattice axes is too high for length-six windows. Length seven is the exact periodic threshold.

For this section only, a *length- k window* is a set of k consecutive cells on one of the three axes. A partial matching is a set of unordered cell pairs in which each cell occurs at most once. A matching *covers* a window if both endpoints of one of its pairs lie in that window.

9.1 The length-six obstruction

The first theorem counts the matched cells demanded separately by the three axis directions. A cell matched along one direction cannot also meet the demand from either of the other two directions.

Theorem 9.1 (No pairing). *There is no partial matching M of the axial hex lattice such that every length-six window contains both cells of a pair in M .*

In plain terms. A position-independent pairing cannot draw Hexo by itself. Each axis needs more than two fifths of the cells to be matched along that axis, while a cell can belong to only one pair. The three demands exceed the available cells. The obstruction concerns a global matching; it does not forbid a partial pairing inside a certified finite region.

Proof. Discard every pair that is contained in no length-six window. This removes all non-axis-aligned pairs and every axis pair whose endpoints have axis distance greater than five. The covering property is unchanged.

Fix one physical axis line and identify its cells with the integers. A pair with endpoints p and $p + \delta$, where $1 \leq \delta \leq 5$, lies in exactly the windows whose start s satisfies

$$p + \delta - 5 \leq s \leq p.$$

It therefore covers $6 - \delta \leq 5$ consecutive window starts. If pairs are ordered by their left endpoints, the intervals of starts covered by consecutive pairs cannot have a gap. Hence

$$p_{i+1} - p_i \leq 6 - \delta_{i+1} \leq 5.$$

A line segment of L cells contains $L - 5$ internal length-six windows and therefore needs at least $(L - 5)/5$ pairs whose endpoints lie in that segment. Those pairs account for at least $2(L - 5)/5$ matched cells along this axis.

Let

$$B_R = \{x : d(x, 0) \leq R\}, \quad |B_R| = 3R^2 + 3R + 1.$$

For any one axis, B_R meets $2R + 1$ line segments, and their lengths sum to $|B_R|$. Summing the preceding lower bound over those segments and then over the three axes gives

$$\text{matched-cell incidences in } B_R \geq \frac{6}{5}|B_R| - 6(2R + 1).$$

Every matched cell contributes to exactly one axis, namely the axis of its pair. A partial matching thus supplies at most $|B_R|$ such incidences. At $R = 20$, the ball has 1261 cells, whereas the lower bound is

$$\frac{6}{5}(1261) - 6(41) = 1267.2 > 1261.$$

This is a contradiction. □

The same density calculation gives a threshold for any translation-invariant line system with negligible boundary. The hypotheses below state the geometric setting needed by the count; they do not refer to an arbitrary graph called a grid.

Corollary 9.2 (Translation-invariant line grids). *Consider k -in-a-row on a point lattice with g pairwise nonparallel, translation-invariant line directions. Suppose that every lattice line is equipped with length- k windows at all integer offsets and that lattice balls have boundary-to-volume ratio tending to zero. If a partial matching covers every length- k window, then*

$$g \frac{2}{k-1} \leq 1, \quad \text{and hence} \quad k \geq 2g + 1.$$

In plain terms. Each direction asymptotically consumes a fraction $2/(k-1)$ of all cells. Boundary cells cannot repair an excessive total because their proportion tends to zero. For the three hex axes the first possible length is seven, so Hexo's target length six falls one cell below the pairing threshold.

Proof. Discard pairs that lie in no target window. On a fixed direction, a pair at axis distance $\delta \in \{1, \dots, k-1\}$ is contained in exactly $k - \delta \leq k-1$ window starts. In a large finite set, all but a boundary term of its internal windows must therefore be covered by at least one pair per $k-1$ window starts. Each such pair uses two cells. The matched-cell density required for that direction is at least $2/(k-1)$ after division by the volume and passage along a boundary-to-volume sequence.

Pairs assigned to distinct directions have disjoint endpoints because the pair family is a matching. Adding the g lower densities gives $2g/(k-1) \leq 1$. Rearrangement gives $k \geq 2g + 1$. □

For comparison, four square-grid directions give the necessary bound $k \geq 9$, consistent with the standard pairing construction for nine in a row. Three hex-grid directions give $k \geq 7$. The next theorem attains that bound on the hex lattice.

9.2 The threshold construction

The construction uses a quotient with twelve cell orbits. Six unit pairs match those orbits, two pairs in each direction. Their lifts select one unit edge phase out of every six on every physical axis line.

Theorem 9.3 (Threshold pairing at $k = 7$). *On the three-axis hex lattice there is a periodic perfect matching such that every length-seven window contains exactly one matched pair. It has a period lattice of index 12, and no periodic construction has smaller index.*

In plain terms. Increasing the target length from six to seven makes a global pairing possible. The twelve-cell tile below matches every cell once and places one pair in every seven-cell window. The index bound says that no smaller periodic tile can do the same job. This theorem locates the threshold; it does not provide a pairing defense for length-six Hexo.

Proof. Write the three positive axis steps as

$$H = (1, 0), \quad V = (0, 1), \quad D = (1, -1).$$

Take the period lattice

$$\Lambda = \langle (2, 2), (0, 6) \rangle = \{(2m, 2m + 6n) : m, n \in \mathbb{Z}\}.$$

Its determinant has absolute value 12, and

$$F = \{0, 1\} \times \{0, 1, 2, 3, 4, 5\}$$

is a fundamental domain. Take every simultaneous Λ -translate of the following six pairs:

$$\begin{array}{ll} H & (0, 0) - (1, 0), \quad (0, 1) - (1, 1), \\ V & (0, 3) - (0, 4), \quad (1, 3) - (1, 4), \\ D & (1, 2) - (2, 1), \quad (1, 5) - (2, 4). \end{array} \tag{9.1}$$

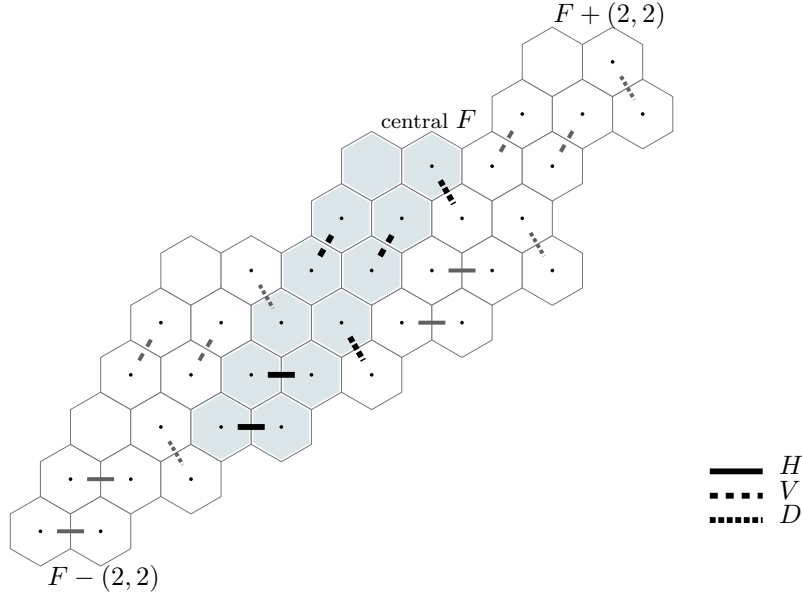


Figure 12: The index-12 pairing tile. The softly shaded cells are the fundamental domain F . The six dark pairs are the representatives in (9.1); lighter pairs are translates by $\pm(2, 2)$. Diagonal representatives cross the displayed fundamental-domain boundary, as required by the quotient matching.

To prove that the lifted pairs form a matching, use the quotient map

$$\phi(q, r) = (q \bmod 2, r - q \bmod 6): \mathbb{Z}^2 \longrightarrow \mathbb{Z}_2 \times \mathbb{Z}_6.$$

Its kernel is Λ . The endpoints in (9.1) have the following images:

$$\begin{array}{ll} H & (0, 0) - (1, 5), \quad (0, 1) - (1, 0), \\ V & (0, 3) - (0, 4), \quad (1, 2) - (1, 3), \\ D & (1, 1) - (0, 5), \quad (1, 4) - (0, 2). \end{array}$$

These are the twelve distinct elements of $\mathbb{Z}_2 \times \mathbb{Z}_6$. Every lattice cell is therefore incident with exactly one lifted pair.

The three axis steps map under ϕ to

$$(1, 5), \quad (0, 1), \quad (1, 4).$$

Each image has order six. Each axis direction consequently forms two six-cycles on the twelve quotient cells. The construction selects one edge on each of those six cycles. Equivalently, the horizontal pair starts on the two line orbits are $(6n, 0)$ and $(6n, 1)$, the vertical starts are $(0, 3 + 6n)$ and $(1, 3 + 6n)$, and both diagonal line orbits have starts with $q \equiv 1 \pmod{6}$. Thus every physical line has one selected unit-edge phase modulo six. A length-seven window has six consecutive internal unit edges, so it contains exactly one selected pair.

It remains to prove periodic minimality. We first record the equality forced on any periodic covering. Pass, if needed, to a period sublattice whose quotient is injective on every set of diameter at most six. Let that quotient have N cell orbits and P pair orbits. Discard pairs lying in no length-seven window. A remaining pair is axis-aligned at distance $\delta \in \{1, \dots, 6\}$ and lies in $7 - \delta \leq 6$ window orbits. There are $3N$ window orbits. Counting pair-window incidences gives

$$3N \leq \sum_e (7 - \delta_e) \leq 6P \leq 3N. \quad (9.2)$$

The last inequality is $2P \leq N$. Equality holds throughout. Hence every cell is matched, every pair is a unit pair, every window contains exactly one pair, and each axis has $N/6$ pair orbits.

On one physical line, let $x_s \in \{0, 1\}$ indicate that the unit edge starting at s is selected. Exact coverage gives

$$\sum_{i=0}^5 x_{s+i} = 1 \quad \text{for every } s.$$

Subtracting the equation at $s + 1$ from the equation at s yields $x_s = x_{s+6}$. Exactly one residue class modulo six is selected, and this indicator has least positive period six.

Table 3: Exact checks for the index-12 pairing.

quantity	result
pair orbits	6
quotient cell orbits checked	12
quotient cell-incidence histogram	$\{1 : 12\}$
quotient start/axis window orbits checked	36
quotient contained-pair histogram	$\{1 : 36\}$
cells checked in $[-30, 30]^2$	3721
finite-patch cell-incidence histogram	$\{1 : 3721\}$
finite-patch windows checked	11163
finite-patch contained-pair histogram	$\{1 : 11163\}$

Return to any period lattice for the same pairing. The order of each axis step in its quotient must be divisible by six, since translation by that order preserves the line indicator. The quotient index N is therefore divisible by six. If $N < 12$, then $N = 6$. An abelian group of order six is cyclic, and its only elements of order six are g and $-g$. The difference of two such elements has order at most three. The third hex-axis step is the difference of the first two, yet it too must have order six. This contradiction proves that index twelve is minimal. $\square$

9.3 Finite verification of the periodic search

The quotient proof reduces the periodic search to six phase variables, one for each of the two line cycles in each of the three directions. Each variable has six possible phases. An exact-cover row consists of one line-cycle constraint and the two endpoint constraints of the selected edge. The resulting instance has 36 rows and 18 columns: six line-cycle columns and twelve cell columns.

The mechanical check exhausts all $6^6 = 46,656$ phase assignments. An MRV Algorithm X traversal visits exactly 419 recursive states and finds 120 labelled solutions. A separate direct traversal of all 46,656 phase tuples also finds 120, including the construction above. The exact structural checks are recorded in Table 3.

The quotient checks are exhaustive rather than samples. Translation by Λ preserves the matching and all three directions, so every cell is equivalent to one of the twelve quotient cells and every window is equivalent to one of the $12 \cdot 3 = 36$ quotient windows. The finite patch is a separate implementation check. The exact-cover encoding, direct enumeration, quotient trace, and patch check are documented with the companion proofs [1].

The two threshold theorems leave a narrow implementation conclusion. A global static pairing is unavailable for Hexo, but partial pairings remain available inside position-dependent certified regions. The index-12 construction applies to the neighboring length-seven game and shows that the density inequality in Corollary 9.2 is attained at $g = 3$.

10 The potential layer and its boundary

Pairing tries to assign one permanent response to every target. A potential argument instead assigns a weight to each surviving target and lets the defender remove as much weight as possible on each placement. The argument is exact for a fixed finite family of windows. On the infinite board, a separate source term is needed when an attacker stone turns a stone-free window into an active target. This distinction determines which conclusions are permanent and which cover only a finite horizon.

Throughout this section the defender is the Breaker in the potential calculation. A real defender completion only ends the game in the defender’s favor, so ignoring that objective is conservative. We use

$$\lambda = \sqrt{3}.$$

This is the largest exponential base supported by the two-against-two greedy comparison of Erdős–Selfridge and Beck [4, 5].

10.1 Fixed target families

Within this section, a window is *potential-active* at P when it is attacker-touched and defender-free: $W \cap A \neq \emptyset$ and $W \cap D = \emptyset$. This is narrower than “alive for A” in Definition 2.6; an all-empty window is not potential-active.

The fixed-family potential retains window labels. Distinct windows continue to count separately even if their residual empty sets happen to coincide. This makes the danger removed by a defender placement an exact sum.

Definition 10.1 (Residual potential and danger). Let F be a finite labelled family of windows. At a position P , let

$$\mathcal{L}_F(P) = \{W \in F : W \cap D = \emptyset\}$$

be its surviving labels and put

$$E_P(W) = W \setminus A.$$

The fixed-family residual potential is

$$\Psi_F(P) = \sum_{W \in \mathcal{L}_F(P)} \lambda^{-|E_P(W)|}. \quad (10.1)$$

For an empty cell x , its fixed-family danger is

$$d_F(x; P) = \sum_{\substack{W \in \mathcal{L}_F(P) \\ x \in E_P(W)}} \lambda^{-|E_P(W)|}. \quad (10.2)$$

The dynamic touched-window potential is

$$\Phi(P) = \sum_{\substack{W: W \cap A \neq \emptyset \\ W \cap D = \emptyset}} \lambda^{-e_P(W)}, \quad e_P(W) = 6 - |W \cap A|. \quad (10.3)$$

Stone-free windows are absent from (10.3).

In plain terms. Each unfinished target receives more weight as the attacker fills it. A defender move deletes every target it meets, while an attacker move increases the terms of targets it advances. The dynamic sum omits untouched windows because the infinite board contains infinitely many of them; assigning each even a fixed positive virgin weight would make the sum diverge.

For a fixed residual family, a defender placement at x decreases Ψ_F by exactly $d_F(x; P)$. An attacker placement at x multiplies each affected term by λ , and therefore increases the potential by exactly $(\lambda - 1)d_F(x; P)$. If all dangers are at most S before an attacker placement, every danger after that placement is at most λS .

Sequential greedy defense recomputes (10.2) before each defender placement and takes an empty cell of maximum danger. Ties follow one fixed deterministic order. If all dangers vanish, the strategy uses a deterministic legal filler. One concrete filler rule chooses an occupied cell with greatest q coordinate, breaks ties by greatest r , and takes its positive Q -neighbor. That neighbor is empty by maximality and is legal at distance one. The rule is recomputed if a second filler is needed.

The next lemma is the numerical core. Its hypothesis about legal positive danger cells holds when every surviving label was potential-active at the initial defender epoch.

Lemma 10.2 (One fixed-family round). *Let the defender make two sequential fixed-family greedy placements, followed by the next attacker turn or its terminal prefix. Suppose every positive-danger cell is legal. If the round reaches the next full defender epoch, the fixed-family potential there is no larger than it was at the initial defender epoch. The corresponding upper bound also holds at a terminal attacker prefix.*

In plain terms. The two greedy defender placements remove at least as much potential as the next attacker pair can restore. Recomputing after the first defender move is part of the argument, because it supplies two separate maxima. The equality $\lambda^2 - 1 = 2$ is the numerical balance behind the result.

Proof. Let δ_1 and δ_2 be the exact reductions on the two defender placements. After both placements, let S be the greatest remaining danger. Deleting labels cannot increase danger at another cell. Each greedy maximum was therefore at least S , so

$$\delta_1 + \delta_2 \geq 2S.$$

The first attacker placement adds at most $(\lambda - 1)S$. The second adds at most $(\lambda - 1)\lambda S$. Their total increase is at most

$$(\lambda - 1)(1 + \lambda)S = (\lambda^2 - 1)S = 2S.$$

The net change is nonpositive. Omitting a terminal second placement can only reduce the increase. $\square$

The strict terminal threshold is one: a completed labelled target has empty residual set and contributes $\lambda^0 = 1$ by itself.

Theorem 10.3 (Fixed-family blocking). *Let P_0 be a nonterminal defender-FirstStone position, and let F be a finite fixed family of windows that are potential-active at P_0 . If*

$$\Psi_F(P_0) < 1, \quad (10.4)$$

then sequential fixed-family greedy defense prevents the attacker from ever completing a member of F .

In plain terms. The theorem protects exactly the targets named at the starting position. It does not automatically enroll a window that is still stone-free there. The full defender turn and the strict inequality are both part of the statement. Fillers are permitted once no positive fixed-family danger remains.

Proof. By Lemma 10.2, induction over full defender epochs keeps $\Psi_F \leq \Psi_F(P_0) < 1$. Defender placements only decrease the sum. If a member of F were completed on the second attacker placement, the next-epoch potential would be at least one, a contradiction. If completion occurred on the first placement, the one-placement version of the round estimate gives

$$\Psi_F \leq \Psi_F(P_t) - 2S + (\lambda - 1)S < 1$$

at that placement, again contradicting its completed term of weight one. Every alleged win occurs at a finite placement, so induction over finite prefixes proves the infinite conclusion. $\square$

The base $\sqrt{3}$ is optimal within this comparison. For a general $\lambda > 1$, the attacker pair can add $(\lambda^2 - 1)S$, while two greedy defender placements guarantee only $2S$. Thus $\lambda \leq \sqrt{3}$ is needed, and the largest permitted base minimizes every unfinished-edge weight. Base 2 is not a substitute: three separated count-four targets have total weight $3 \cdot 2^{-2} = 3/4 < 1$, but the defender kills at most two of them. The inequality in (10.4) also cannot be relaxed to equality at the correct base. Three pairwise separated count-four targets have total weight $3\lambda^{-2} = 1$ and give the same loss.

The theorem is sufficient, not necessary. A single defender placement can kill many labelled targets sharing its cell even when their total weight is at least one. The full-turn premise also matters. With only one defender placement remaining, two separated count-four targets have $\Psi_F = 2/3 < 1$; the defender kills one and the attacker completes the other.

10.2 Birth mass and permanent local certificates

The family in Theorem 10.3 is fixed. Dynamic touched-window greedy enrolls targets during play, and their first contribution is not paid by the fixed-family round estimate. The exact missing quantity is their birth mass.

Let P_t be the start of a full defender turn. During the following attacker pair, let N_t contain the windows that were stone-free at P_t , survived the intervening defender pair, and first became potential-active during that attacker pair. For $W \in N_t$, let $a_t(W) \in \{1, 2\}$ be the number of stones from the pair that lie in W at the end of the pair or terminal prefix. Define

$$C_t = \sum_{W \in N_t} \lambda^{-(6-a_t(W))}. \quad (10.5)$$

If $u_{1,t}$ windows are born with one stone and $u_{2,t}$ with two, then

$$C_t = \frac{u_{1,t}}{9\sqrt{3}} + \frac{u_{2,t}}{9}.$$

The initial potential-active cohort obeys the fixed-family round estimate, while the new cohort contributes exactly (10.5). Hence every completed round satisfies

$$\Phi(P_{t+1}) \leq \Phi(P_t) + C_t. \quad (10.6)$$

Theorem 10.4 (Cumulative-birth certificate). *Let P_0 be a nonterminal defender-FirstStone position. Fix the tie-breaking and filler policy above and use sequential dynamic touched-window greedy from P_0 . Suppose there is a constant B_∞ such that, uniformly over every legal attacker strategy τ ,*

$$\sup_\tau \sum_{t=0}^{\infty} C_t(\tau) \leq B_\infty, \quad \Phi(P_0) + B_\infty < 1. \quad (10.7)$$

Then the defender prevents every attacker completion. More generally, if

$$\sup_\tau \sum_{t=0}^{T-1} C_t(\tau) \leq B_T, \quad \Phi(P_0) + B_T < 1, \quad (10.8)$$

then the first T attacker pairs are safe.

In plain terms. New targets may be enrolled, but their total initial weight must be reserved in advance. The bound is uniform over attacker strategies; a small birth sum observed on one line is not a certificate. The theorem is useful when a finite region or another argument supplies a finite activation reservoir.

Proof. Before attacker pair t , repeated use of (10.6) gives

$$\Phi(P_t) \leq \Phi(P_0) + \sum_{i=0}^{t-1} C_i < 1.$$

Apply the one-cycle result proved in Theorem 10.6 below. It makes pair t safe, after which (10.6) advances the bound. Induction proves (10.8). Under (10.7), the same induction applies to every finite prefix produced against every τ . $\square$

A finite confinement gives a concrete reserve. Fix a nonterminal defender-*FirstStone* position P_0 and a finite set R of cells empty at P_0 . Suppose every future attacker placement lies in R . Put

$$\begin{aligned} \mathcal{A}_R &= \{W : W \cap D_0 = \emptyset, W \cap A_0 \neq \emptyset, W \setminus A_0 \subseteq R\}, \\ \mathcal{V}_R &= \{W : W \text{ is stone-free at } P_0, W \subseteq R\}, \end{aligned} \tag{10.9}$$

and let

$$\Phi_R(P_0) = \sum_{W \in \mathcal{A}_R} \lambda^{-e_{P_0}(W)}, \quad N_R = |\mathcal{V}_R|.$$

For a matching M of vertex-disjoint two-cell subsets of R , let $q(M)$ be the number of windows in $\mathcal{V}_R$ containing both cells of some pair in M , and define

$$P_R = \max_M q(M), \quad B_R^* = N_R \lambda^{-5} + P_R(\lambda^{-4} - \lambda^{-5}). \tag{10.10}$$

Theorem 10.5 (Finite-region certificate). *Assume the attacker is forever confined to the finite set R . If*

$$\Phi_R(P_0) + B_R^* < 1, \tag{10.11}$$

then delayed-enrollment region-target greedy prevents every attacker win. The weaker condition

$$\Phi_R(P_0) + \frac{N_R}{9} < 1 \tag{10.12}$$

is also sufficient. Every positive-reduction defender placement is an empty cell of a currently potential-active member of $\mathcal{A}_R \cup \mathcal{V}_R$ and is legal. Fillers need not lie in R .

In plain terms. Confinement converts infinitely many possible virgin targets into a finite list. Each such target pays a one-stone birth charge, with an extra charge if both of its first attacker stones arrive in one turn. The theorem does not prove confinement; a solver must verify that premise by another certificate.

Proof sketch. Every virgin target is enrolled once, with charge λ^{-5} , or λ^{-4} when both first stones arrive in one attacker turn. Those exceptional birth pairs form a matching in R , so at most P_R targets receive the larger charge and the total is at most $B_R^* \leq N_R \lambda^{-4} = N_R/9$. For the potential Ξ_R of the currently potential-active members of $\mathcal{A}_R \cup \mathcal{V}_R$, delayed greedy gives $\Xi_R(P_{t+1}) \leq \Xi_R(P_t) + C_t^R$ and the charge gives $\sum_t C_t^R \leq B_R^*$. The induction of Theorem 10.4 therefore keeps $\Xi_R < 1$. Confinement places every possible completed window in this target family, and every positive-danger cell is a legal empty within distance five of an attacker stone. The complete charging argument is in the companion [1]. $\square$

For checking (10.11), P_R must be exact or replaced by a proved upper bound. The elementary bound

$$P_R \leq \min\{N_R, 5\lfloor |R|/2 \rfloor\}$$

follows because two distinct cells lie together in at most five length-six windows. One exhibited matching gives only a lower bound on P_R and cannot be inserted into (10.10) as a certificate.

10.3 Finite-horizon consequences

Even without a birth reserve, a stone-free length-six window cannot be filled in one attacker pair. This gives the immediate dynamic result.

Theorem 10.6 (One-cycle certificate). *If the defender is at FirstStone and $\Phi(P_t) < 1$, then two sequential dynamic touched-window greedy placements prevent an attacker win during the immediately following two-placement turn, including its first-placement win check.*

In plain terms. The current active targets are below the fixed-family threshold, so greedy blocks them through the next attacker turn. A target that is still stone-free cannot acquire six stones in two placements. The potential may exceed one after those placements, so this is not a permanent certificate.

Proof. Apply Theorem 10.3 to the potential-active windows at P_t . A window outside that cohort either contains a permanent defender stone or is stone-free at P_t . The latter receives at most two attacker stones during the turn and cannot be completed. $\square$

Fixing the initial cohort gives a longer horizon because a virgin target needs six attacker placements, regardless of where the placements are split into turns.

Theorem 10.7 (Five-placement certificate). *If the defender is at FirstStone and $\Phi(P_0) < 1$, then the defender has a strategy preventing an attacker win during the next five attacker placements: two complete attacker turns and the first placement of the third.*

In plain terms. The defender uses fixed-family greedy on the targets active at the start. Those targets stay blocked forever. Every other live candidate began empty and still lacks at least one attacker stone after five future placements. The sixth placement is outside this guarantee.

Proof. Let F be the potential-active family at P_0 and use fixed- F sequential greedy. The initial potential of F is $\Phi(P_0) < 1$, so Theorem 10.3 blocks it forever. A window containing a defender stone at P_0 remains dead. Every remaining window was stone-free and has at most five attacker stones after five future attacker placements. $\square$

The three-pair result uses a deliberate defender placement against virgin targets and only one greedy placement on the same turn. The following lemma gives the exact cost of that split.

Lemma 10.8 (One-greedy amplification). *Let F be a fixed potential-active family at a full defender epoch and write $X = \Psi_F$. Suppose a defender turn consists of one arbitrary legal placement followed by one F -greedy placement, and then at most two attacker placements. At every attacker prefix, including a terminal placement,*

$$\Psi_F \leq \frac{3}{2}X. \quad (10.13)$$

If both attacker placements are nonterminal, this is also the bound at the next defender epoch. If $3X/2 < 1$, no member of F is completed at either placement.

In plain terms. One defender placement may be spent on a virgin target rather than on fixed-family danger. The remaining greedy move still limits the fixed-family growth to a factor of three halves for that round. This controlled loss of margin permits the three-pair strategy to mix geometric blocking with potential defense.

Proof. The arbitrary defender placement leaves a potential $X' \leq X$. Let the greedy placement reduce it by δ , and let S be the greatest remaining danger. Deletion monotonicity and greediness give $S \leq \delta$. A cell's danger is also a subsum of the remaining potential, so $S \leq X' - \delta$. The next attacker pair adds at most $2S$ by the same calculation as Lemma 10.2. Therefore

$$\Psi_F^+ \leq X' - \delta + 2 \min\{\delta, X' - \delta\} \leq \frac{3}{2}X' \leq \frac{3}{2}X.$$

After only the first attacker placement, its increase is at most $(\lambda - 1)S \leq 2S$, so the same bound holds. A completed member contributes one and contradicts (10.13) when $3X/2 < 1$. $\square$

The geometry of the first attacker pair identifies every virgin window that could be completed on the sixth future attacker placement. Blocking those windows costs zero, one, or two split defender turns according to the pair's axis relation.

Theorem 10.9 (Three-pair certificate). *Let P_0 be a nonterminal defender-FirstStone position, and let x, y be the first future attacker pair. The defender can prevent a win during the next three complete attacker turns under the respective conditions*

$$\Phi(P_0) < \begin{cases} 1, & x, y \text{ lie in no common window,} \\ 2/3, & x, y \text{ have common-window axis distance 2, 3, 4, or 5,} \\ 4/9, & x, y \text{ are axis-adjacent.} \end{cases} \quad (10.14)$$

In particular, $\Phi(P_0) < 4/9$ unconditionally certifies all three attacker turns.

In plain terms. A virgin window can first win on the sixth future attacker placement. Its first two attacker cells reveal which virgin windows could contain all six placements. The defender blocks that small common family while retaining enough fixed-family potential margin. Adjacent first cells expose five possible windows and require two split defender turns, which accounts for the smaller threshold $4/9$.

Table 4: One-window contributions to the dynamic potential.

attacker stones	empties	exact weight	decimal
1	5	$1/(9\sqrt{3})$	0.064150
2	4	$1/9$	0.111111
3	3	$1/(3\sqrt{3})$	0.192450
4	2	$1/3$	0.333333
5	1	$1/\sqrt{3}$	0.577350

Proof sketch. Let F be the initial potential-active family and $X_t = \Psi_F(P_t)$; the first greedy defender pair gives $X_1 \leq X_0$, while a virgin sixth-placement completion must contain both x and y . If they share no window, later F -greedy pairs suffice. At common-window distance 2 through 5, write $y = x + dv$ for a signed unit axis vector v ; playing $x + v$ and then one greedy cell hits every eligible virgin target and gives $X_2 \leq 3X_1/2 < 1$. When x, y are adjacent, playing $x - v$ leaves at most $W_* = \{x, x + v, \dots, x + 5v\}$, which a later deliberate hit blocks, and two one-greedy estimates give $X_3 \leq (3/2)^2 X_1 \leq 9X_0/4 < 1$. Thus the initial family remains below threshold and every eligible virgin target is hit. The companion gives the legality and filler cases in full [1]. $\square$

10.4 How narrow is the raw threshold?

For $s \in \{1, \dots, 5\}$ let n_s count potential-active windows with exactly s attacker stones. Then

$$\Phi = \sum_{s=1}^5 n_s \lambda^{s-6},$$

with the weights shown in Table 4.

If $I = \sum_{s=1}^5 s n_s$ is the number of live attacker-stone-window incidences, direct substitution gives $\lambda^{s-6}/s \geq 1/18$, with equality at $s = 2$. Thus

$$\Phi \geq I/18,$$

and $\Phi < 1$ forces $I \leq 17$. Since each attacker stone has eighteen window incidences, the raw threshold occurs only after almost all such incidences have been killed by defender stones. It is not a generic opening or middlegame condition.

The comparison can be made without floating point. Set

$$a = n_1 + 3n_3 + 9n_5, \quad b = n_2 + 3n_4.$$

Then $\Phi < 1$ holds exactly when

$$b \leq 8 \quad \text{and} \quad a^2 < 3(9 - b)^2. \quad (10.15)$$

Indeed, after multiplication by $9\sqrt{3}$, the inequality is $a < \sqrt{3}(9 - b)$. Its right side must be positive, after which squaring preserves the comparison.

10.5 Dynamic greedy can be forced to lose

The fixed-family theorem does not prove that repeatedly recomputing the dynamic touched family is a global defense. The next finite counterexample branches over every exact greedy maximum. All maxima are positive, so the result is independent of the filler policy.

Theorem 10.10 (All-ties greedy refutation). *Let*

$$A_0 = \{(0, 0)\}, \quad D_0 = \{(1, 0)\},$$

with the defender at FirstStone. Then $\Phi(P_0) < 1$. Against every sequential dynamic touched-window greedy defense, the fixed attacker continuation

$$\begin{array}{ll} \text{turn 1} & (2, -4), (2, 2), \\ \text{turn 2} & (-5, 0), (-4, 0), \\ \text{turn 3} & (-3, 0), (-2, 0), \\ \text{turn 4} & (-1, 0) \end{array} \quad (10.16)$$

wins on the first placement of turn 4. The completed window is $\{(-5, 0), (-4, 0), \dots, (0, 0)\}$.

Table 5: Exhaustive dynamic-greedy branch counts. Danger values are scaled by 27.

defender placement	input boards	exact maximum distribution	output boards
$D0.1$	1	$5\sqrt{3}$	4
$D0.2$	4	$5\sqrt{3}$ (4)	4
$D1.1$	4	$6\sqrt{3}$ (2), $7\sqrt{3}$ (2)	12
$D1.2$	12	$5\sqrt{3}$ (12)	124
$D2.1$	124	$10\sqrt{3}$ (124)	242
$D2.2$	242	$9\sqrt{3}$ (6), $10\sqrt{3}$ (236)	124
$D3.1$	124	$21 + 4\sqrt{3}$ (124)	124
$D3.2$	124	$3 + 9\sqrt{3}$ (106), $10\sqrt{3}$ (18)	124

In plain terms. Greedy is not rescued by choosing ties more carefully. The verifier branches over every maximum-danger reply and reaches the same fixed winning placement on all 124 surviving boards. This refutes the dynamic greedy strategy, not the claim that some different defender strategy may survive from every position with $\Phi < 1$.

Machine-verified finite proof. The defender stone (1,0) kills five of the eighteen windows through (0,0). The initial profile is therefore $(n_1, n_2, n_3, n_4, n_5) = (13, 0, 0, 0, 0)$, and

$$\Phi(P_0) = \frac{13\sqrt{3}}{27} = \frac{13}{9\sqrt{3}} < 1,$$

because $13^2 = 169 < 243 = 3 \cdot 9^2$. Every attacker placement in (10.16) is legal on every branch. The first two are at distance four from (0,0), and each later placement is within distance five of an attacker stone on the target line.

The verifier represents every potential and danger as an integer pair (a, b) denoting $(a + b\sqrt{3})/27$. Comparisons use integer arithmetic. When the coefficients of a difference have opposite signs, the verifier compares a^2 with $3b^2$; irrationality excludes equality. It enumerates every maximum at each defender placement, merges only identical colored boards at the same phase, and then applies the fixed attacker continuation. The exact state counts and maxima are shown in Table 5. There is no random choice, beam limit, or depth cutoff.

The last defender turn contains the trap. Immediately before $D3.1$, the cell $(-6, 0)$ lies in five potential-active horizontal windows with attacker counts 1, 2, 3, 4, 4. Hence

$$27d(-6, 0) = \sqrt{3} + 3 + 3\sqrt{3} + 9 + 9 = 21 + 4\sqrt{3}. \quad (10.17)$$

The winning cell $(-1, 0)$ lies in one count-four window and the count-five target, so

$$27d(-1, 0) = 9 + 9\sqrt{3}. \quad (10.18)$$

Their difference is $12 - 5\sqrt{3} > 0$, since $12^2 > 3 \cdot 5^2$. Thus every branch chooses $(-6, 0)$ uniquely. That stone kills the count-four term at $(-1, 0)$ and leaves

$$27d(-1, 0) = 9\sqrt{3}.$$

On 106 boards the unique second maximum is $(-3, 1)$ with danger $3 + 9\sqrt{3}$. On the other 18 it is $(-2, -1)$ with danger $10\sqrt{3}$. Both exceed $9\sqrt{3}$. No greedy branch occupies $(-1, 0)$.

The verifier also checks radius-eight legality, the absence of every attacker or defender completion before the terminal placement, every potential-active-window profile, every exact maximizer, and the terminal window. At the terminal placement all 124 boards have exactly one completed attacker window and hence $\Phi \geq 1$. The scheduled attacker cell is absent from every prior maximizer union. This exhausts every dynamic-greedy tie-breaking sequence and proves the theorem. The exact checker and its full profile trace are included with the companion proofs [1]. $\square$

The compact position has unequal material counts, but the same tree occurs in a legal engine history.

Proposition 10.11 (Reachable greedy witness). *There is a legal 39-placement history ending at a nonterminal defender-FirstStone position with 20 attacker stones, 19 defender stones, the same initial profile $(13, 0, 0, 0, 0)$, and exactly a translated copy of the greedy tree in Theorem 10.10.*

Proof sketch. Translate the compact core by $(-1, 0)$, so its defender stone is the opening at (0,0) and its attacker stone is $(-1, 0)$. Let

$$c = (-11, 10), \quad I = B_2(c), \quad R = B_3(c) \setminus B_2(c), \quad g = (-9, 8).$$

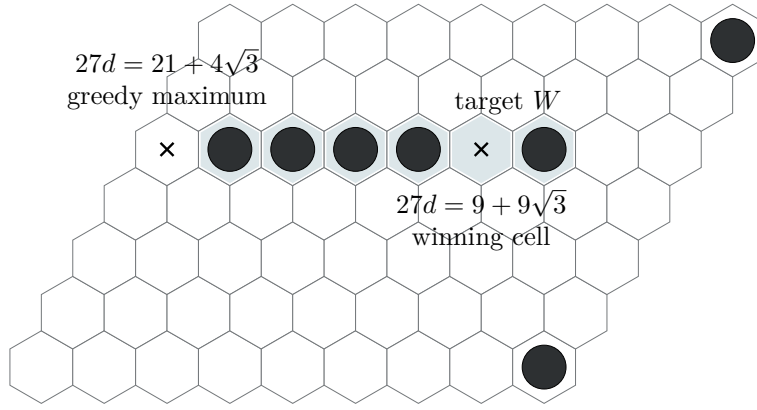


Figure 13: The common attacker geometry on all 124 boards immediately before $D3.1$. The attacker set is $\{(-5, 0), (-4, 0), (-3, 0), (-2, 0), (0, 0), (2, -4), (2, 2)\}$. Defender stones are omitted because their locations depend on the greedy branch. The target, candidate cells, and displayed danger values are the same on every branch.

Here $|I| = 19$ and $|R| = 18$; after the opening, the attacker takes $(-1, 0)$ and g , then nine cycles alternate two unused cells of R with two unused cells of $I \setminus \{g\}$. All moves are legal because $d(g, (-1, 0)) = 8$ and every cell of $B_3(c)$ is within five of g . No prefix has a six-run, and the final defender ring blocks every window meeting I while distance-six padding isolates the translated tactical core. The count is $1 + 2 + 9 \cdot 4 = 39$ placements with material totals 20 and 19. The companion checker verifies every prefix and reruns the full greedy tree [1]. $\square$

10.6 Clean escape and failure of raw renewal

The greedy counterexample concerns one strategy. A separate geometric lemma shows that no strategy can keep the raw sublevel set $\Phi < 1$ invariant at every defender epoch. It still does not give an attacker win against every non-greedy defense.

Lemma 10.12 (Universal clean escape). *From every finite, nonempty, nonterminal position immediately before an attacker turn, the attacker has a legal pair that births exactly 36 distinct one-stone windows. Its birth mass is*

$$36\lambda^{-5} = \frac{4}{\sqrt{3}}. \quad (10.19)$$

In plain terms. The attacker can move twice toward an unoccupied extreme of the finite board. Both local stars are clean and share no target window. The resulting birth mass already exceeds one, so raw current potential cannot renew itself as an epoch-by-epoch invariant under any defender policy.

Proof. Put $h(q, r) = q + r$. Choose an occupied cell z maximizing h , and write $M = h(z)$. Play

$$x = z + (4, 4), \quad y = x + (4, 4).$$

Both displacements have hex distance eight, so the placements are legal in sequence. They are empty because their h -values exceed M .

A horizontal or vertical length-six window through x has minimum h at least $h(x) - 5 = M + 3$. A diagonal window through x has constant $h = M + 8$. Every such window was therefore stone-free. A horizontal or vertical window through y has minimum h at least $M + 11$, while a diagonal window through y has constant $h = M + 16$. Those windows were stone-free even after the placement at x . Finally, $y - x = (4, 4)$ lies on none of the three axes, so x and y share no window. Each placement births its eighteen incident windows, all with one attacker stone. The 36 distinct terms have total weight

$$36(\sqrt{3})^{-5} = \frac{4}{\sqrt{3}}.$$

$\square$

Starting from one defender stone and no attacker stone, $\Phi = 0$. After any defender pair, Lemma 10.12 makes the next defender-epoch potential at least $4/\sqrt{3} > 1$. Repeating the construction on every continuing attacker turn creates 36 fresh labels each time. The cumulative birth sum diverges, even though each such label may remain at count one or be killed. Radius-eight mobility alone therefore supplies no summable reserve for Theorem 10.4.

10.7 Two static repair attempts

The first no-go result strengthens [Theorem 9.1](#). Finite occupation cannot remove enough exceptional windows to make a static matching possible on the remaining virgin board.

Proposition 10.13 (Finite saturation does not enable static pairing). *No partial matching covers all but finitely many length-six windows. Hence no finite initial position admits a static pairing that covers every initially virgin target.*

Proof. Suppose a matching misses at most K windows. Discard its pairs that lie in no length-six window. Let B_R be the radius- R hex ball, with

$$|B_R| = 3R^2 + 3R + 1.$$

For each axis and $R \geq 5$, the $2R + 1$ line segments through B_R contain

$$|B_R| - 5(2R + 1)$$

length-six windows wholly inside the ball. Across three axes, at least

$$3|B_R| - 15(2R + 1) - K$$

of these internal windows must be covered. One matched pair is contained in at most five windows on its axis. The ball therefore contains at least

$$\frac{2}{5}(3|B_R| - 15(2R + 1) - K)$$

matched-cell incidences. A matching supplies at most $|B_R|$, so

$$|B_R| \leq 60R + 30 + 2K.$$

This contradicts $|B_R| = 3R^2 + 3R + 1$ for sufficiently large R .

If a finite position has occupied support S , at most $18|S|$ windows meet that support. Every other window is initially virgin. A matching covering all virgin windows would miss only finitely many windows, contrary to the first part. $\square$

The second no-go result concerns a static additive account that tries to put a summable, spatially damped weight on every virgin window while retaining the same factor-three two-placement growth comparison.

Proposition 10.14 (Static two-tier damping no-go). *Suppose a nonnegative static per-window account assigns weight $w_s(W)$ when W contains s attacker stones. Assume*

$$\sum_W w_0(W) < \infty, \quad w_6(W) \geq 1,$$

and, for every window,

$$w_{s+2}(W) \leq 3w_s(W), \quad s = 0, 2, 4. \quad (10.20)$$

No such account exists.

Proof. For every window, the terminal bound and the three growth inequalities give

$$1 \leq w_6(W) \leq 3w_4(W) \leq 9w_2(W) \leq 27w_0(W).$$

Thus $w_0(W) \geq 1/27$ for each of infinitely many windows. Their virgin sum cannot be finite. $\square$

The proposition leaves other accounting systems open. Its hypotheses do not cover age-dependent weights, moving centers, cancellation between windows, a nonadditive account, or an adaptive pairing whose ownership changes during play.

10.8 Finite reduction and the remaining boundary

Although the unrestricted question has no known scalar renewal, every fixed horizon occupies a finite part of the ideal board. Compactness then relates all finite survival policies to one infinite policy.

Theorem 10.15 (Finite-horizon equivalence). *Fix a finite nonempty position P with occupied support S_0 of size n , at defender-FirstStone. For $h \geq 1$, define*

$$R_h = \{x : d(x, S_0) \leq 32h\}. \quad (10.21)$$

Every move in the first h full defender/attacker rounds lies in R_h , and

$$|R_h| \leq n(3072h^2 + 96h + 1). \quad (10.22)$$

The defender blocks forever from P if and only if the defender has a survival strategy for every finite h -round game on R_h . The set R_h is a proved move universe through that horizon, not an added board boundary.

In plain terms. Every fixed-depth game can be searched exactly on a finite board region. If a survival policy exists at every depth, König’s lemma selects policies whose finite restrictions agree and joins them into one infinite strategy. The theorem gives no depth after which survival automatically implies permanent safety; checking every horizon restates the infinite problem rather than solving it algorithmically.

Proof. The first h rounds contain at most $4h$ placements. Induct on the future placement index k . A legal placement lies within distance eight of an occupied cell, so the k th placement lies within distance $8k$ of S_0 . Since $k \leq 4h$, every move lies in R_h .

A radius- r hex ball contains $3r^2 + 3r + 1$ cells. The union bound over the n initial support cells with $r = 32h$ gives

$$|R_h| \leq n(3(32h)^2 + 3(32h) + 1) = n(3072h^2 + 96h + 1).$$

The truncated legal game tree is finite, so backward induction decides it.

One implication is immediate: a forever-surviving strategy restricts to a survival strategy for each horizon. For the converse, form a rooted tree of finite defender policy tables. Level h contains the policies that survive the h -round game, and an edge joins a policy to its restriction at level $h - 1$. Every level is finite, every node has finitely many extensions, and every level is nonempty by hypothesis. König’s lemma supplies an infinite chain of compatible policies. Their union is one defender strategy that survives every finite prefix and hence blocks forever. Conversely, if no forever strategy exists, some finite level is empty; finite minimax then gives an attacker strategy forcing completion within that horizon. $\square$

The results above leave five separate proof obligations. They should not be collapsed into one generic statement that the potential method is incomplete.

GAP-RAW. Theorem 10.10 defeats every dynamic touched-window greedy tie-breaking, but it gives no attacker strategy against arbitrary non-greedy play. No theorem here proves that a non-greedy forever defense always exists. The raw existential statement remains open.

GAP-GLOBAL-RENEWAL. Theorems 10.7 and 10.9 extend the certified horizon, but neither returns to a defender epoch satisfying the same raw hypothesis. Lemma 10.12 proves that no defense can use $\Phi < 1$ itself as the renewed invariant, even when the starting value is zero.

GAP-AMORTIZED-ABANDONMENT. An online account must discount indefinitely many weak births that the attacker abandons, yet remain sound if dormant stones later participate in multi-axis forks. Clean escape produces a positive exact birth debit with no maturity promotion. Proposition 10.14 excludes only the stated static factor-three accounts and supplies no history-sensitive refund rule.

GAP-DYNAMIC-PAIRING. Proposition 10.13 excludes a static matching even after finite initial exceptions. It does not exclude an adaptive response system whose pairs or ownership assignments change with the play.

GAP-FINITE-CUTOFF. Theorem 10.15 proves that every loss has a finite witness and that each fixed horizon is decidable. It gives no computable or uniform bound $H(P)$ such that survival through $H(P)$ rounds implies survival forever.

The strategy-level nonclaims also affect implementation. A raw node-local $\Phi < 1$ test may certify Theorem 10.6 or seed one of the stated finite-horizon strategies. It must not be converted into an unrestricted defender win, draw, or global move-set pruning rule. Every positive greedy move lies in an empty cell of a currently potential-active monitored window and is legal. If no such target remains, one or both mandatory placements can be fillers. A solver implementing a potential-certified strategy must retain a legal filler fallback, and it must account for the filler’s effect on the future radius-eight legality frontier. Full derivations and the exact finite traces appear in the companion proof documents [1].

11 Mechanized validation

The main soundness results are mathematical proofs. Some auxiliary claims also have finite machine checks. Other computations measure a search generator or test a local rule on sampled positions. These roles are different. This section records the input class, the published count, and the conclusion supported by each run. The scripts and full traces are described in the repository companion documents [1].

Table 6: Published search-harness counts and their scope.

Run	Published count	Supported conclusion
Local-distance sample	400 positions; maxima 2 and 3	Confirms the predicted local distances on the sample only.
Radius-two junction	2,374 nodes; 217 restricted turns	Gives a concrete false-pruning control for radius two.
Random radius-two comparison	109 positions; 0 genuine divergences	Measures rarity in the sample; does not restore soundness.
Closure junction	1,396 uncapped nodes	Confirms that the tested closure retains the known junction defense.
Closure random comparison	91 positions; 52 claims; 0 divergences; 139 seconds	Tests the generator at matched horizons; does not prove ranked-zone coverage.

11.1 Search-harness runs

The compact search model reproduces the engine’s coordinates, legality, phase machine, per-placement terminal check, and threat masks. Random replay found matching legal sets and terminal decisions after each placement. No aggregate concordance count is published, so this remains implementation evidence rather than an exhaustive equivalence proof.

On 400 random positions, the largest observed distances were two for an immediate completion or threat hit and three for threat creation. A separate universe measurement found means of about 302 legal cells, 121 radius-three cells, and 80 radius-two cells. The radius-three plus attacker-touched predecessor generator later measured about 147 cells against the roughly 302-cell legal baseline. These are measurements, not verifier constants.

The engineered junction position supplies a negative control. Restricting the defender to the radius-two set produced an attacker win after 2,374 uncapped search nodes. The run proved all 217 turns of the form “mandatory pin hit plus radius-two spare” losing under that restriction. The omitted distance-three junction reply, paired with the same mandatory hit, was not refuted by an uncapped search through depth eight. The comparison establishes that radius two is an unsound defender restriction. It does not establish that the junction reply is a full-game defender win. The defender-side survival probe stops at the first unrefuted child, which is valid for non-loss at the tested horizon but not for a LOSS proof.

A comparison on 109 attackable positions found no genuine radius-two versus full-set divergence. One node-cap artifact was re-adjudicated at two million nodes and gave the same WIN verdict in both variants. A sparse failure mode still invalidates a universal pruning rule.

On the engineered junction, the predecessor closure contained the junction cell and the defender survived after 1,396 uncapped nodes. A bounded run on 91 attackable positions produced 52 restricted WIN claims, zero full-set divergences at the matched horizon, and a time of 139 seconds. The radius-two controls show that the harness can expose an unsound restriction. The closure run remains search evidence; it does not prove Theorem 6.5 or replace Theorems 5.1 and 6.1.

11.2 Domination spot checks

The domination rules in Section 6 have analytic proofs. Two finite randomized checks exercise their most error-prone rule channels. The turn-commutation check generated 2,997 legal same-mover pairs after filtering out every singleton-winning prefix. It found zero differences in the final occupancy, masks, legal support, next phase, or terminal winner. The dead-cell check generated 400 adversarial configurations. The stones that make the focal cell dead were placed at large permitted offsets. It found zero violations of the radius-eight frontier containment in Lemma 6.11. The worst cell in the focal radius-eight ball was at distance six from an existing stone, leaving a distance margin of two.

These checks are not exhaustive proofs of the domination patterns. The companion gives finite quotient specifications for MV-P1, MV-P2, and MV-P3, including 289 marked local geometries for MV-P1 and 81 for MV-P3 after the stated local-incidence quotient. It also counts 942,779,391,290 dead-star colourings up to the dihedral symmetry in the MV-P1 regression space. Those numbers describe prospective enumeration spaces. The full quotient enumerations have not been run. In particular, the 2,997-pair and 400-position checks do not justify dropping the frontier-inertness premise of P1, the dead-spoke and support premises of P2, or either singleton premise of P3.

11.3 Four finite-check campaigns

Four focused campaigns accompany the forced-hit, tightness, pairing, and potential results. The review log records that the repaired assertion-enabled checkers exit successfully. Only the pairing and potential reports publish exhaustive state counts. The tightness and forced-hit reports publish the checked predicates but no aggregate assertion count.

Threshold pairing. The pairing checker works on the twelve-cell quotient of the period lattice in Theorem 9.3. Its exact-cover instance has 36 rows and 18 columns: six line-cycle constraints and twelve endpoint constraints. Direct phase space has $6^6 = 46,656$ assignments. An MRV Algorithm X traversal exhausted the instance in 419 recursive states and found 120 labelled solutions. A separate direct iteration over all 46,656 phase tuples also found 120. The displayed six-pair construction occurs in both enumerations.

The checker verifies six pair orbits and twelve quotient cell orbits. The cell-incidence histogram is $\{1 : 12\}$. It verifies 36 quotient start-and-axis window orbits, with contained-pair histogram $\{1 : 36\}$. As a redundant finite-patch test, it checks all 3,721 cells in $[-30, 30]^2$ and all 11,163 length-seven windows starting there. The two patch histograms are $\{1 : 3721\}$ and $\{1 : 11163\}$. Translation by the period lattice makes the quotient checks exhaustive for the infinite construction. This computation establishes the stated matching and coverage properties and exhausts the six phase variables for that quotient. The index-minimality argument is analytic. The computation does not produce a pairing for length six, does not replace the density proof of Theorem 9.1, and does not rule out position-dependent partial pairings.

Exact-surd greedy refutation. The potential checker represents each value as an integer pair for $(a + b\sqrt{3})/27$ and compares pairs without floating point. It includes every exact greedy maximizer along the fixed attacker continuation in Theorem 10.10. The successive input/output state counts for the eight defender placements are

$$1/4, \quad 4/4, \quad 4/12, \quad 12/124, \quad 124/242, \quad 242/124, \quad 124/124, \quad 124/124.$$

State deduplication merges only identical coloured boards at the same phase. At the last defender placement, 106 states have unique second maximum $(-3, 1)$ and 18 have unique second maximum $(-2, -1)$. All 124 branches omit the attacker’s winning cell. The checker verifies legality, every nonterminal prefix, the exact alive-window profiles, every maximum and tie, and one terminal window on all 124 branches. It also constructs a legal 39-placement engine history for the enlarged reachable witness and replays the same full greedy tree there.

There is no randomness, beam, or depth cutoff in this enumeration. It proves that the named fixed attacker continuation defeats every dynamic-greedy tie-breaking on the stated position. It does not prove that every defender strategy loses from every position with $\Phi < 1$. That raw existential question remains open. It also does not weaken the fixed-family and finite-region sufficiency theorems, whose target families do not renew in the same way.

Tightness checks. The tightness checker publishes the following summary. The triangle has three threats, transversal number two, and minimum witness size three. The five-cycle has five threats, transversal number three, and minimum witness size five. The T6 diagnostic has $K_1 = \{k\}$ while the omitted cell is an immediate defender completion. The OR-COMPLETION and leaf-entry deadline roles pass their protection checks. The exact-rank, virgin-radius, touched-equality, and T5 local distance/count traces pass. The report then states that all tightness checks passed. It publishes no aggregate number of assertions or configurations.

The checker establishes the listed finite mask, transversal, deadline, and arithmetic facts. The remaining coordinate families and framework-scope arguments in the tightness report are prose proofs. In particular, a checked equality for one fixed window does not create a complete countercertificate for the full union of zone terms. The OPEN and relative classifications in Section 12 therefore remain part of each result.

Forced-hit review. The forced-hit report gives an eight-stage finite verifier procedure. It recomputes each named threat mask and empty set; checks the transversal number and the defender non-win diagnostic; enumerates the kernel; checks exact successors and the ban on kernel substitution; propagates checkpoint roles; computes the least f and queried Q^D values in reverse topological order; enumerates the four debited zone terms at ordinary nodes; and checks every escape deadline. The 15-plus-15 sharpness position separately checks all 18 incident windows per stone, nonterminality, and the stated maximum defender count. The report and review log record a successful assertion-enabled checker run, but they publish no aggregate count of assertions, queried windows, or certificate nodes.

The machine run is a consistency check on finite data and recurrences. The proof of Theorem 7.8 is analytic and has status PROVEN, not PROVEN-MECH. The run does not prove a target-independent reduction of B , a zero-cost substituted hit, automatic net shrinkage of the final searched set, or a debit for slack pressure. Those nonclaims are part of the PARTIAL boundary of the gate calculus.

11.4 Validation boundary

The validation layers support different statements. Exhaustive quotient enumeration proves the finite pairing construction. Exact branch enumeration proves the all-ties greedy counterexample. The tightness and forced-hit checkers confirm named finite calculations while their main claims remain analytic or scope-qualified. Randomized domination checks test implementations of proven local rules. Search-harness runs measure a candidate generator and supply negative controls. None of these runs allows a verifier to accept an omitted defender reply without the hypotheses of the theorem used at that node.

12 Tightness: the limit map

The constants in the certificate have several kinds of boundary. An *absolute pin* is witnessed by a weakened certificate that declares an attacker win although a legal defender line defeats the declared horizon. A *relative pin* attains a counting bound or breaks the stated coupling, clock, kernel, or witness representation. It leaves open a verifier with additional state. A *syntactic pin* is required to make the certificate grammar and its recurrences defined. An *arithmetic attainment* reaches equality for one selected window or local trace but does not supply a complete certificate in which every other zone term is absent. OPEN means that no general pin and no smaller sound replacement is known. PARTIAL means that a restricted reduction is proved and broader forms remain open.

The tables give the current value and one reason for every row. The reason column states the scope of the boundary. A value marked relative must not be reported as a game-wide optimum. A value marked absolute must retain the combined premises named in that row.

12.1 Obligation and completion boundaries

Rows R1a and R9 describe one mechanism. A real-only legality chain can advance by eight at each ordinary defender opportunity. The coordinate family attains $8(r - 1)$ before a rank- r carrier is checked. Lowering the step or exact-rank band breaks the first-protected-occupation argument. The uniform replacement $8(B - 1)$ is different. Its soundness follows from $r \leq B$, but no complete terminal certificate realizes all required data with $r = B$. Its sharpness remains open.

Rows R2 and R3 distinguish one selected window from the verifier's full union. A virgin-window trace uses $E - 6$ distance-eight relays and six fills, attaining $8(E - 6)$. An incident virgin window can still select the same seed at distance zero, so this does not pin Z_{virgin} . A touched window with two common defender stones and four real fills attains $2 + 4 = 6$. This pins the local comparison but lacks the recurrence labels, terminal data, and exclusion of other zone terms needed for a full countercertificate.

12.2 LOSS and kernel boundaries

Every threat window has one or two empties. At defender budget one, the triangle family

$$\{a, b\}, \quad \{a, c\}, \quad \{b, c\}$$

has transversal number two and needs all three members. At budget two, the five-cycle has transversal number three and deleting any edge drops it to two. A six-member minimal obstruction would be K_4 . It cannot arise from Hexo windows: every empty pair is axis-collinear, and four such cells either violate the three-axis incidence geometry or lie on one line, where an intervening empty prevents a required two-cell mask. The three/five caps are exact for the named-window format, not for every possible witness language.

Rows R5a and R5c are limits of the exact-kernel grammar. The scope $\text{mhs}(P) \leq b$ keeps the kernel nonempty, and $b - 1$ is the placement capacity after the current reply. R5b is absolute only under combined enforcement: if both the T6 premise and D9 diagnostic omit the defender non-win test, a singleton kernel can exclude an immediate defender completion. Keeping either test rejects that certificate.

R6 has the same combined-clause scope. At $\tau(\mathcal{T}) = b$, the defender can hit every named witness. This defeats a contract that accepts equality. If the universal survivor clause is still checked directly, removing the redundant numeric test has no effect.

12.3 Clock, deadline, and grammar boundaries

The R8 and R12 scalar recurrences are attained path maxima. An AND edge adds one, a LOSS remainder may use all b placements, and an OR edge adds none. Window exposure stops on attacker entry. Indexed clocks may still be smaller for one role or window. The D17 transition charge in R7 is needed because the real reply can legalize the next relay or add a window fill before the child clock starts. Another envelope may record that event differently, but it cannot omit the event.

R10 mixes relative rank clauses with absolute deadline clauses. AND +1, OR +0, and the maximum over roles are limits of the parent-wide union coupling. Dropping protection before leaf entry lets the real defender kill a LOSS witness that remains alive in the ghost. Omitting an OR-COMPLETION role lets the real defender occupy the designated winning cell. Both admit a false resolution.

R11 is absolute for a separate reason. Among $b + 1$ disjoint count-four windows, the defender kills b and leaves one with two empties. The second following attacker placement completes it, attaining leaf-ply plus $b + 2$. R15 is only syntactic: a zero-child AND supplies no coupling filler and leaves child maxima undefined, but deleting nonemptiness alone gives no false-win witness.

Table 7: Limit map, obligation and completion rows.

ID	Parameter and value	Status	One-line reason
R1a	Exact live-role band $8(r - 1)$	Relatively pinned	A distance-eight relay chain reaches the protected carrier after exactly r defender opportunities, so a smaller band breaks $L9'$.
R1b	Uniform live-role band $8(B - 1)$	OPEN	The exact-rank chain does not supply a complete certificate with $r = B$ and synchronized terminal, legality, and LOSS data.
R2	Virgin radius $8(E^D - 6)$ for $E^D \geq 6$	Fixed-window arithmetic attained; full union OPEN	A relay of $E^D - 6$ steps attains the radius for one window, but one of the seed's other incident virgin windows may still select it.
R3	Touched guard $\text{cnt}_D(W) + E^D(W) \geq 6$	Fixed-window equality attained; full pin OPEN	Two existing stones followed by four real fills attain equality, but no complete recurrence-labelled certificate excludes every other mandatory zone.

Table 8: Limit map, LOSS and kernel rows.

ID	Parameter and value	Status	One-line reason
R4a	LOSS witness cap at $b = 1$: three windows	Relatively pinned	The three triangle pairs have transversal number two, while every two-window subfamily has a one-cell transversal.
R4b	LOSS witness cap at $b = 2$: five windows	Hexo-specific reduction; five relatively pinned	The five-cycle needs all five edges for transversal number three, and hex-window geometry excludes the generic six-edge K_4 obstruction.
R5a	Internal T6 scope $\text{mhs}(P) \leq b$	Relatively pinned	At transversal number $b + 1$ the exact kernel is empty, so it cannot be the searched set of a nonempty ordinary internal node.
R5b	Kernel enforcement of $\neg \text{own_win_now}(P)$	Absolute for combined enforcement only	If both the T6 premise and the D9 diagnostic are omitted, the singleton kernel can exclude an immediate defender-winning reply.
R5c	Residual test $\tau(F \setminus d) \leq b - 1$	Relatively pinned	A first hit of two disjoint singleton threats leaves the one remaining placement required by the exact minimum-transversal line.
R6	LOSS contract $\tau(\mathcal{T}) > b$ plus universal survivor	Absolute for the combined contract	At equality the defender can hit every named witness; the numeric test alone is redundant when the universal survivor clause is still checked.

Table 9: Limit map, clocks and grammar rows.

ID	Parameter and value	Status	One-line reason
R7	D17 transition charge: $1 + \text{child rank and exposure}$	Relatively pinned	The current real-only reply can be the missing legality relay or the missing window fill, so child-only labels omit one causal event.
R8	D14 budget: $\text{AND } 1 + \text{max, LOSS } b$	Exact as defined; uniform use relative	A maximizing finite path uses the current edge and every LOSS remainder placement, although indexed guards may use smaller quantities.
R9	Legality-chain coefficient eight	Relatively pinned	The rules legalize a relay at distance exactly eight, and the protected occupation and virgin traces attain that step length.
R10	D15: $\text{AND } +1, \text{OR } +0, \text{deadline zero, role maximum, protection through check}$	Mixed; leaf-entry and OR-COMPLETION clauses absolute	Rank and maximum clauses are limits of the union coupling; dropping either named deadline protection admits a concrete false certificate.
R11	LOSS deadline $p_{\text{leaf}} + b + 2$	Absolutely pinned	After b hits, a surviving count-four window has two empties and completes only on the second following attacker placement.
R12	D16: $\text{AND } 1 + \text{max, LOSS } b, \text{stopping value zero}$	Exact as defined; uniform use relative	The recurrence is the attained path maximum before resolution or first attacker entry, while a branch-indexed hazard may be smaller.

12.4 Static coverage and the forced-hit boundary

R13 attains radius three at budget three and gives local failure at budget four, where a touched empty at distance four satisfies $2 + 4 = 6$. Neither trace supplies the full certificate and clock data for an absolute T5 pin. R14 is relative for a different reason. An explicit fourth future attacker placement creates a threat supported only by future stones while the cell was initially outside the touched-window set and radius three.

At a protected tight gate, an exact copied hit adds no real-only defender stone. The f clock can omit that opportunity, while Q^D charges it only for a selected window containing the hit and charges full escape and LOSS remainders. The scalar B , full r , and full E^D remain. A D17 substitute may create a frontier, checkpoint roles may enlarge the protected set, and slack pressure leaves a quiet placement. R16 therefore proves hazard debits, not scalar reduction, free substitution, automatic net shrinkage, or a slack-pressure rule.

The limit map has three practical readings. The absolute rows are verifier safety boundaries under their stated combined premises. The relative rows identify where this certificate representation uses all of its available count. The OPEN and PARTIAL rows identify changes that require a new proof, not a smaller constant inserted into the current formulas.

13 Conclusions

13.1 What this paper establishes, in plain language

Hexo gives a player two stone placements on an ordinary turn. A line of six wins as soon as its last stone is placed. The board is unbounded, but a legal placement must lie within distance eight of an existing stone. These rules create a search problem with a local tactical goal and a broad legal reply set. A midgame defender node may have hundreds of legal cells. Checking all of them at every level is often the main cost of a threat-space search.

The basic result is a safe way to omit some of those cells. The omission is not justified by distance alone. It is justified by a finite certificate. The certificate records the attacker's planned placements, the empties needed by terminal threat witnesses, the number of defender opportunities before each use, and the windows in which a hidden defender stone could complete a line. A verifier forms a zone from those records. Direct obligations are searched. A future obligation that is not yet legal receives a seed band wide enough to cover every possible radius-eight relay. Defender-alive windows receive separate guards according to whether they already contain a defender stone or are still empty.

The dismissal theorem says that a certificate passing these checks survives every legal defender reply, including replies outside the searched set. Its proof follows a real game and a ghost game together. The real game contains the omitted reply. The ghost game contains a searched filler and follows the certificate. The zone prevents the first omitted stone from occupying a future required cell or completing a defender window. The attacker therefore reaches the same certified resolution, or a permitted adaptive LOSS resolution, by the declared placement horizon. Search trimming can still miss an attacker win. It cannot turn a losing attack into an accepted win when the verifier enforces the certificate conditions.

The zone can be computed with cell-specific deadlines and window-specific exposures. This is no larger than assigning every object the full certificate depth, and can be smaller. A tight current threat family admits a different reduction. At a protected forcing gate, an exact copied defender hit does not create a real-only legality frontier. The debited clocks remove that opportunity from the relevant seed band and charge it to a selected window only when the hit lies in that window. A defender that refuses the gate is handled by an explicit escape contract. This result is conditional on exact copying and the gate checkpoint. It does not reduce the global scalar budget and does not make a substituted hit free.

Terminal LOSS certificates also have a bounded description. With one defender placement left, three threat windows suffice. With two placements left, five suffice. The triangle and five-cycle families show that these caps are exact for the named-window witness format. The result reduces proof data without changing the universal survivor condition.

Two global defenses have exact boundaries. A position-independent pairing cannot cover every length-six window on the three-axis hex lattice. At length seven, a periodic perfect matching exists, covers every window exactly once, and has minimum period index twelve among periodic constructions. The potential method gives sound fixed-family, finite-horizon, and finite-region blocking certificates at base $\sqrt{3}$. It does not give a global dynamic greedy defense: an exact 124-branch enumeration gives a fixed attacker line that defeats every greedy tie-breaking from a position with $\Phi < 1$. A non-greedy global defense from every such position is not ruled out.

The resulting rule is concrete. A solver may skip a reply only with a checked reason tied to the selected proof. Exact-copy gates, three/five LOSS witnesses, fixed-family potential bounds, and the narrow domination patterns provide such reasons under their premises. Distance-only pruning, unrestricted dynamic greedy, and a global pairing for length six do not.

13.2 Results and limits

Table 11 separates the result consumed by a verifier from its main limit. The detailed quantitative boundary is the limit map in Section 12.

Rule constants such as window length six, eighteen incident windows, and the radius-eight legality step are not tuning

Table 10: Limit map, static coverage and forced-hit rows.

ID	Parameter and value	Status	One-line reason
R13	T5 static cover: $B \leq 3$ with r_3	Local arithmetic attained; full pin OPEN	Radius three is attained at $B = 3$ and the local formula fails at $B = 4$, but the coordinate trace is not a complete D9 certificate.
R14	L10 cutoff: first three future threat-creating placements	Relatively pinned	An explicit fourth placement creates a count-four virgin target while lying outside both the initial touched-window set and r_3 .
R15	Independently nonempty AND set: at least one legal reply	Relative and syntactic	A zero-child AND supplies no ghost filler and leaves D14/D16 maxima undefined, but deleting nonemptiness alone gives no false-win witness.
R16	Forced-hit debit: exact-copy f and Q^D ; full B, r, E^D retained	PARTIALLY RESOLVED	Tight protected gates debit copied hits for named hazards; scalar, substituted-hit, slack-pressure, and net-shrinkage claims remain open.

Table 11: Results and their operative limits.

Layer	Established result	Operative limit
Dismissal	A valid zone certificate preserves the attacker win against every legal defender reply by its horizon.	The result is certificate-relative and may lose search completeness.
Ranked zones	Exact role ranks and per-window exposures determine finite direct, seed, touched, and virgin terms.	Uniform-band and full-union virgin sharpness remain open.
Forced-hit gates	Protected exact-copy gates permit f and Q^D hazard debits.	No scalar B debit, free D17 substitution, automatic net shrinkage, or slack-pressure debit is proved.
LOSS witnesses	Three windows suffice at $b = 1$ and five at $b = 2$.	The lower examples pin the named-window witness representation.
Pairing	No global covering matching exists for $k = 6$; a minimal-index periodic construction exists for $k = 7$.	Position-dependent partial pairings are not excluded.
Potential	Fixed-family and finite-region inequalities certify blocking; finite horizons admit explicit bounds.	Dynamic touched-window greedy fails globally; the raw non-greedy claim is open.
Domination	P1, P2, and P3 permit checked local quotienting.	Their frontier, dead-spoke, initial-legality, and singleton-nonwin premises are mandatory.
Machine checks	Pairing and the greedy counterexample have exhaustive finite enumerations; other runs test named calculations or samples.	Search experiments and randomized spot checks are not theorem proofs.

parameters. Within the certificate format, $8(r - 1)$, the LOSS deadline $p_{\text{leaf}} + b + 2$, the three/five witness caps, and the leaf-entry and OR-COMPLETION deadlines have attained or absolute boundaries. Scalar path maxima, transition charges, and exact-kernel tests are instead framework-relative; indexed fields or another grammar may encode the same hazards differently.

The uniform $8(B - 1)$ wrapper and full-union virgin radius remain open. The gate calculus is partial: it debits exact copied hits from named hazards, but an escape, LOSS remainder, or substitution retains the full clock. Added checkpoint roles may offset a smaller radius. Neither a fixed-window equality nor a hazard debit promises a smaller total set at every node.

13.3 Implications for solvers

A df-pn implementation can separate search from verification. Search may use any move ordering or candidate generator. It may start with the measured radius-three and attacker-touched predecessor set, use threat incidence for ordering, or learn a proposal distribution. When that search finds a candidate win, a compiler attaches the finite grammar, exact child map, obligations, clocks, window data, and terminal witnesses. An independent verifier then checks the theorem selected at each defender node. Failure of one check yields an unverified or unknown result, not an attacker win.

At an ordinary node, the verifier recomputes legality and masks, then checks protected cells, ranks, window exposures, searched-set containment, attacker legality, terminal data, and the horizon. A DAG also needs one consistent position, phase, label, and clock assignment at each shared node. Branch-indexed substitution uses its transition-inclusive envelope.

Deep certificates gain most from exact labels. A uniform band uses the largest remaining defender budget around every future carrier. A ranked band uses only the defender opportunities before that carrier's own deadline. Windows that the attacker enters early stop their exposure clocks. A DAG can share repeated suffixes without increasing the selected path budget. These reductions target the part of the proof where a coarse horizon creates the largest spatial bands.

Forcing lines admit an additional node type. The verifier first checks the named current threat masks, $\tau(F_Q) = b$, and the absence of an immediate defender win. It protects every threat empty through that checkpoint. It then searches exactly the extendable-hit kernel. A kernel reply is copied at the same cell and cannot use D17 substitution. A reply outside the kernel abandons the old subtree and invokes the finite escape contract with deadline $p(Q) + b + 2$. Ordinary descendants may use the debited clocks. A loose threat family with $\tau(F_Q) < b$ does not qualify for this reduction.

At a LOSS leaf, the compiler may retain at most three or five windows. The verifier still checks their masks, completion legality, universal survivor condition, and defender non-win diagnostic. The cap does not validate a heuristic threat list.

No fixed partner table covers length six. Potential checks apply to their fixed-family or finite-region premises; dynamic danger is otherwise an ordering heuristic. P1, P2, and P3 also retain their frontier, dead-spoke, initial-legality, and singleton-nonwin checks. Ordered intermediate nodes remain when a certificate depends on them.

The measured 147-cell predecessor set against roughly 302 legal cells is a generator target, not a theorem zone. Learned proposals and node caps have the same status. They may shorten search, while only an accepted verifier rule reduces the defender set in the proof.

13.4 Future work

Broader gate debits. The forced-hit result leaves four connected questions. A target-independent scalar budget smaller than B would shrink every coarse wrapper, but it must cover terminal ordering, LOSS remainders, and every window at once. A substituted hit can change the legality frontier even when it answers the same threat, so a sound D17 debit needs a stronger domination or frontier-inertness certificate. Checkpoint roles may offset the saved hazard radius, and a tight-gate theorem alone does not prove net searched-set shrinkage. A family with transversal number below the placement budget also leaves a quiet move; a finite residual-state account for that slack could extend the gate method. Resolving these questions would determine when forcing pressure produces a predictable end-to-end branch reduction.

Uniform and full-union geometry. The uniform $8(B - 1)$ obligation wrapper may contain slack not present in the exact ranks. A smaller bound would simplify certificates that do not carry cell-specific deadlines. It requires either a complete equality family or a role-and-turn theorem showing that a ghost-illegal carrier always has less than the full budget. The full-union virgin radius has a different obstacle: overlap among the eighteen incident windows can select a seed that the target window no longer selects. A solution would either construct certificates excluding every competing term or prove that overlap always yields a smaller safe union.

Potential boundary. Five questions remain distinct. GAP-RAW asks whether every nonterminal defender-FirstStone position with $\Phi < 1$ has some non-greedy forever defense. GAP-GLOBAL-RENEWAL asks for an invariant that absorbs the stream of new touched windows. GAP-AMORTIZED-ABANDONMENT asks whether killed or abandoned targets can fund that renewal without a divergent virgin sum. GAP-DYNAMIC-PAIRING asks whether pairs chosen with

the position can complement the potential account even though a static global pairing is impossible. GAP-FINITE-CUTOFF asks for an a priori horizon bound for a finite witness to failure or survival. A solution to the last question would turn the König-style finite reduction into a bounded decision procedure.

Domination and local enumeration. The P1 and P2 premises are deliberately narrow. A richer causal boundary state could track unmatched frontier chains and live mask differences for a fixed horizon. Such a state may certify more dead-cell replacements or more interchangeable hits. It must preserve early terminality, actor-directed legal inclusion, and every crossing window. Running the full MV-P1, MV-P2, and MV-P3 quotient specifications would add independent exhaustive checks of the present local patterns. It would not by itself prove a broader pattern.

Formalization. The certificate objects are finite. Positions have finite occupancy, windows queried by a certificate form finite families, transversal tests have budgets at most two, and the tree or DAG is finite. These features make a Lean or Coq reference checker feasible. A practical order is to formalize axial distance and windows, the phase machine and terminal order, the D9 grammar, and the scalar recurrences. The real-ghost coupling, ranked first-bad-event proof, transition envelopes, and gate escape proof can then be layered on those definitions.

The work is substantial. The unbounded board requires careful finite-support lemmas. LOSS leaves quantify over a finite but adaptive defender remainder. DAG sharing needs path clocks and label consistency. Gate checkpoints add a two-phase protected set and a separate escape horizon. Potential boundary results use exact algebraic comparisons and a compactness argument. A formalization would give a small trusted verifier and expose omitted scope conditions, but it would not certify the search heuristics or settle the OPEN rows without new mathematics.

A Notation and source crosswalk

The notation tables collect the symbols used across the certificate, pairing, potential, and domination layers. The crosswalk then records the paper number and the corresponding tag in the repository companion. Core tags without a prefix refer to *PROOF_TSS_DEFENDER_ZONES.md*. The prefix ES-P refers to *ES_POTENTIAL.md*, ES-G refers to *ES_GLOBAL_BOUNDARY.md*, and DOM refers to *DOMINATION.md*. The crosswalk has no row for expository prose or an unnumbered observation.

A.1 Notation

Table 12: Game and certificate notation.

Symbol	Meaning
$\mathbb{Z}^2 = \mathbb{Z}^2$	Axial cell lattice.
$x = (q, r)$	Cell with axial coordinates q, r .
$d(x, y)$	Hex distance $\max\{ q - q' , r - r' , (q - q') + (r - r') \}$.
$B_R(x), B_R(U)$	Closed radius- R ball around a cell, and the union of those balls around a set.
$\mathbf{a}_q, \mathbf{a}_r, \mathbf{a}_s$	The three unoriented axes represented by $(1, 0)$, $(0, 1)$, and $(1, -1)$.
W	Six consecutive cells on one axis.
$\Omega(x)$	The eighteen windows containing x .
$P = (\sigma, m, b)$	Finite position, player to place, and remaining placements in the current turn.
$\text{Stones}(P)$	Occupied cells of P .
$\text{Legal}(P)$	Legal empty placement cells of a nonterminal position.
$\text{cnt}_X(W, P)$	Number of stones of player $X \in \{A, D\}$ in W .
$E(W, P)$	Empty cells of W in P .
$\text{own_win_now}(P)$	Predicate that the mover can complete a window within the remaining placement budget.
$\text{hit}(P)$	Family of empty sets of current opponent-threat windows.
$\text{mhs}(P), \tau(F)$	Minimum transversal size of the current threat family, or of a finite set family F .
p	Absolute placement index on a certificate path.
T	Declared absolute resolution horizon.
N, P_N	Certificate node and its labelled position.
$S(N)$	Legal defender replies represented by children of an AND node.
$B(N)$	Local upper bound on defender placements before certified resolution, including a LOSS remainder.
$\Omega(N)$	Live obligation roles reachable below N .
$\text{Prot}(N)$	Cells carrying live attacker-placement or leaf-witness roles at N .
$r_N(\rho), r_N(y)$	Remaining ordinary defender opportunities before role ρ is checked, and the maximum rank carried by cell y .
$E_N^D(W)$	Maximum defender exposure before attacker resolution or first attacker entry into W .

Table 13: Zone, coupling, and gate notation.

Symbol	Meaning
$Z_{\text{dir}}(N)$	Legal cells carrying a current direct obligation.
$Z_{\text{seed}}(N)$	Legal seeds close enough to relay into a future protected carrier before its deadline.
$Z_{\text{touch}}(N)$	Empties of touched defender-alive windows whose current count plus exposure can reach six.
$Z_{\text{virgin}}(N)$	Legal seeds that can reach and fill an all-empty defender window within its exposure.
$R_{\text{cert}}(\mathcal{C}, N)$	Mandatory certificate-relative union of the applicable zone terms.
$R_{\text{search}}(\mathcal{C}, N)$	Optional search superset obtained by adding heuristic candidate terms.
X, Y	Real-only dismissed defender stones and ghost-only searched filler stones.
$\hat{X}$	History of cells that have ever entered the real-only difference set.
$K_b(N)$	Extendable-hit kernel at remaining defender budget b .
$\hat{B}, \hat{r}, \hat{E}$	Transition-inclusive D17 envelope labels for a selected child.
$\mathcal{G}$	Finite acyclic certificate DAG.
H_Q	Named attacker-threat windows at forcing gate Q .
$F_Q = \{E(U, P_Q) : U \in H_Q\}$	Gate threat-empty family.
$K(Q)$	Exact-copy gate kernel $\{d : \tau(F_Q \setminus d) \leq b - 1\}$.
(Q, U, e)	Checkpoint role protecting empty e of named gate window U through the entry mask check.
$\text{Prot}^-(Q), \text{Prot}^+(Q)$	Protected set immediately before and after checkpoint discharge at a gate.
$f_N(\rho), f_N(y)$	Forced-hit-debited ordinary-opportunity clock for a role and the maximum such clock carried by a cell.
$Q_N^D(W)$	Branch-coherent maximum of full-cost defender opportunities plus exact copied gate hits lying in W .
$Z_{\text{dir}}^{FH}, Z_{\text{seed}}^{FH}$	Direct and seed terms at an ordinary debited node.
$Z_{\text{touch}}^{FH}, Z_{\text{virgin}}^{FH}$	Touched and virgin completion terms using Q^D .
$p(Q) + b + 2$	Gate escape deadline after entry with b placements left.

Table 14: Pairing, potential, and domination notation.

Symbol	Meaning
M	Partial matching used by a pairing strategy.
$\Lambda = \langle (2, 2), (0, 6) \rangle$	Index-twelve period lattice of the length-seven pairing construction.
$\phi(q, r) = (q \bmod 2, r - q \bmod 6)$	Quotient map with kernel Λ .
$\lambda = \sqrt{3}$	Base of the Hexo potential.
$\Phi(P)$	Sum of $\lambda^{-e(W)}$ over attacker-touched alive windows in the dynamic family.
$\Psi_F(P)$	Residual potential of a fixed finite target family F .
$d_P(x)$	Potential reduction, or danger, of a defender placement at x .
C_t	Birth mass enrolled after an attacker pair at epoch t .
Φ_R, B_R^*, N_R	Finite-region potential, refined activation reserve, and coarse number of region targets.
$(n_1, \dots, n_5)$	Counts of attacker-alive windows containing one through five attacker stones.
$\Lambda(P)$	Geometric legal support $\bigcup_{s \in \text{Stones}(P)} B_8(s)$ in the domination layer.
Out_n	Stopped outcome through n placements after a compared defender reply.
$b \preceq_n a$	Defender reply b is n -outcome-dominated by searched reply a .
G, H	Searched and omitted states in a causal alternating simulation.

A.2 Paper number to source tag

The first column uses live references so the displayed paper numbers follow the final section order. Each table below has exactly the paper-number and source-tag columns.

Table 15: Core definitions and the geometric fact.

Paper number	Source tag
Definition 2.1	D1
Definition 2.2	D2
Definition 2.4	D3
Definition 2.5	D4
Definition 2.6	D5
Definition 2.7	D6
Definition 2.8	D7
Definition 2.9	D8
Definition 4.1	D9
Definition 4.3	D10
Definition 4.8	D11
Definition 4.10	D12
Definition 6.4	D13
Definition 4.2	D14
Definition 4.4	D15
Definition 4.5	D16
Definition 6.6	D17
Definition 6.9	D18
Definition 7.1	D19
Definition 7.2	D20
Definition 7.3	D20a
Definition 7.4	D21
Fact 2.3	F1

Table 16: Core lemmas and results.

Paper number	Source tag
Lemma 3.1	L1
Lemma 3.2	L2
Lemma 3.5	L3
Lemma 3.6	L4
Lemma 3.7	L5
Lemma 3.8	L6
Lemma 3.9	L7
Lemma 4.9	L9'
Lemma 4.6	L10
Lemma 4.7	L11
Lemma 4.11	L12
Lemma 8.1	L13
Lemma 6.8	L14
Lemma 7.5	L15
Lemma 7.6	L16
Lemma 7.7	L17
Theorem 3.3	T1
Theorem 3.4	T2
Theorem 5.1	T3
Theorem 6.1	T4
Theorem 6.2	T5
Theorem 6.3	T6

Table 16 continued

Paper number	Source tag
Theorem 6.5	T7
Theorem 9.1	T8
Corollary 9.2	T8.1
Theorem 9.3	T8.2
Theorem 6.7	T9
Theorem 6.10	T10
Theorem 7.8	T11
Theorem 7.9	T11.1

Table 17: Potential- and domination-layer results.

Paper number	Source tag
Definition 10.1	ES-P D3–D4
Lemma 10.2	ES-P L5
Theorem 10.3	ES-P T1
Theorem 10.6	ES-P T2
Theorem 10.4	ES-P T3
Theorem 10.5	ES-P T4
Theorem 10.10	ES-G T1
Proposition 10.11	ES-G P1
Lemma 10.12	ES-G L1
Theorem 10.7	ES-G T2
Lemma 10.8	ES-G L2
Theorem 10.9	ES-G T3
Proposition 10.13	ES-G P2
Proposition 10.14	ES-G P3
Theorem 10.15	ES-G T4
Lemma 6.11	DOM L7
Proposition 6.12	DOM P1
Proposition 6.13	DOM P2
Proposition 6.14	DOM P3

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